How Wrong Is a Simulation Prediction, and Why? Residual-Based Quantification and Hierarchical Attribution

Yujing Lin
Supply Chain Optimization Technologies, Amazon, Austin, TX, USA
Jingtao Zhang
Supply Chain Optimization Technologies, Amazon, Seattle, WA, USA
Yunan Liu
Supply Chain Optimization Technologies, Amazon, New York, NY, USA
Corresponding author: jingtaz@amazon.com

Abstract

Stochastic simulation is routinely used to forecast system performance and to evaluate policies that have never been implemented, yet the uncertainty reported with a prediction is often only Monte Carlo sampling error, even when imperfect inputs, limited model fidelity, and inherent system noise dominate the gap between simulation and reality. We develop a residual-based framework organized around three questions: *How wrong is the prediction? What kind of error is it?* and *Why does it arise?* Observable residuals give a learnable target for total prediction error even when only one realized outcome is available per period, provided error is modeled on the scale of the ratio of realized to predicted output: in a strongly seasonal, heavy-tailed system neither series is poolable across periods, but their ratio is. After a separate calibration model, which we consume rather than develop, suppresses systematic bias, a first decomposition separates intrinsic uncertainty, reducible by additional replications, from extrinsic uncertainty, which is not. A second attributes that uncertainty to individual inputs by Shapley value, for a single run or as it changes between two runs, with a correction that preserves efficiency when calibrated means differ. Both decompositions act on the same estimand, so the three answers account for one quantity rather than standing as separate analyses, and skewness detection turns predicted error into calibrated quantiles and prediction intervals. On a production-scale retail supply chain simulator, nominal 80% intervals based on Monte Carlo error alone attain 7.7% empirical coverage against 77.7% when the full uncertainty is included, and in a counterfactual lead-time experiment the extrinsic component changes the significance conclusion in 6 of 17 planning weeks while attribution identifies the inputs responsible. The result is a diagnostic for choosing among more computation, better inputs, and a different model, rather than a single error bar.

Keywords: stochastic simulation; input uncertainty; uncertainty quantification; variance decomposition; Shapley value; prediction intervals; counterfactual analysis; supply chain

1 Introduction

Stochastic simulation is used for two tasks that are central to operational decision-making: predicting how a complex system will perform in the future and evaluating how that system would behave under a policy that has never been implemented. In both cases the simulation produces a number (a forecast, a quantile, a confidence interval, or a treatment effect) that may feed a costly decision. The natural question is therefore not only what the simulator predicts, but *how wrong that prediction may be*.

That question is harder than it appears. A stochastic simulation contains randomness generated internally by finite Monte Carlo replication, but its inputs are themselves estimated from data, its representation of the real system is imperfect, and the physical system remains stochastic even under a perfect model. Consequently, a narrow Monte Carlo confidence interval need not imply a precise prediction of reality. It may only imply that the simulator has estimated its own conditional mean precisely.

This distinction is particularly important in large operational simulations. Such models are attractive precisely because dependence is generated endogenously by the system logic. For example, in a supply chain simulator a labor constraint at one warehouse can create backlog across many products simultaneously; the resulting dependence arises from the modeled operations rather than from an externally specified covariance matrix. The same simulation also preserves physical consistency across aggregation levels. These properties make simulation valuable for planning, but they also make its inputs numerous, dependent, and difficult to estimate accurately. More replications can reduce Monte Carlo error without addressing any of those deficiencies.

The operating environment also limits how error can be measured. A prediction is made for a future period, and that period occurs once: the realized outcome is a single draw, and reality cannot be replicated the way a simulation can. Nor are successive periods repeated trials of one experiment. Demand is strongly seasonal, holiday and promotional peaks move the level by more than the spread within either regime, and the distribution of outcomes carries a heavy right tail. The inputs are re-estimated every period and the operating configuration itself changes, so even the mapping from inputs to outcome drifts. Pooling history therefore does not straightforwardly yield a sample of how wrong the simulator is: a pooled set of raw discrepancies mixes regimes rather than describing any one period. Both the sources of error and the difficulty of observing them shape what a useful answer looks like.

We therefore organize simulation prediction error around three questions: *How wrong is the prediction? What kind of error is it? Why does it arise?* The first asks for a total risk measure on the scale relevant to a future realization. The second asks whether that error is intrinsic to finite simulation or extrinsic to the simulation run itself. The third asks which estimated inputs are responsible for the reducible extrinsic component. These questions form a hierarchy because the answer to each determines whether the next question is worth asking.

To make the hierarchy precise, consider four mechanisms that separate a simulation prediction from reality:

1. *Monte Carlo sampling.* A finite number of replications leaves sampling error in the simulated mean. This component is reducible by computation.

2. *Input error.* Input distributions are estimated from finite real-world data and can be wrong in both location and shape. Their effect does not disappear when the number of simulation replications grows.

3. *Model fidelity gaps.* The simulator necessarily omits or approximates parts of the real system, creating systematic discrepancy between simulated and real behavior.

4. *Inherent system noise.* Even under perfect inputs and perfect fidelity, a future realization fluctuates around its own mean. This component is irreducible for prediction.

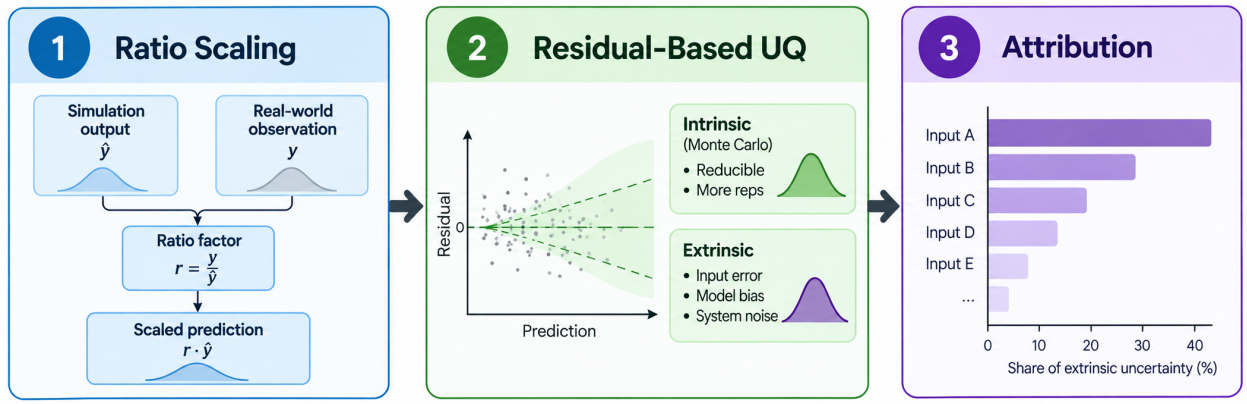


Figure 1: End-to-end uncertainty quantification and attribution framework. The framework proceeds in three stages: (1) *ratio scaling* places prediction error on the scale of the ratio of realized to predicted output, the scale on which error is poolable across periods; (2) *residual-based uncertainty quantification* estimates total prediction error and separates intrinsic Monte Carlo uncertainty from the remaining extrinsic component; and (3) *attribution* uses Shapley values to identify the inputs driving extrinsic uncertainty. The stages are three readings of one quantity rather than three separate analyses, and their order is the order of the interventions available to a decision-maker.

The difficulty is that these components are not separately observed. In particular, the true system mean is unavailable: each period provides a realized outcome, not repeated observations of the counterfactual world needed to estimate that mean. Our approach therefore starts from the quantity that *is* observable, the residual between the realized outcome and the simulation-based prediction. We learn the conditional second moment of that residual as a measure of total prediction error.

The scale on which that residual is measured matters more than it appears, for the reason sketched above: each period yields a single residual, so the second moment has to be learned by pooling across periods, and raw residuals do not pool. The simulator's own output does not pool either, since it inherits the same seasonality. Their ratio does, because the simulator is an explicit model of the structure that makes periods differ, so dividing the realized outcome by the simulated prediction removes most of that structure. Modelling error on this ratio scale is what turns one observation per period into a usable sample, and it is what later makes counterfactual error prediction possible at all; Section 3.2 makes the argument in full.

Systematic bias is suppressed by a calibration model that this paper does not develop. The calibrated mean is consumed as an input: a separate model supplies it, as described in Section 3, and the framework below would accept any center that is approximately unbiased. Stating that division of labour early avoids a misreading, because the center is not incidental to what we report. A more accurate center tightens the quantiles and intervals of Section 3.3, and a drifting one would undermine the premise that lets the second moment of the residual be read as uncertainty rather than as squared bias, a dependence we return to among the limitations in Section 6. What this paper contributes is the uncertainty around the center, not the center itself. Conditional on that calibration, total uncertainty is decomposed into an intrinsic component estimated from simulation replications and an extrinsic residual, and Shapley values then allocate the extrinsic component across input drivers. Figure 1 shows the resulting sequence.

What ties the stages together is that they are not three analyses but three readings of one quantity. The predicted error, its split into intrinsic and extrinsic parts, the per-input attribution, and the uncertainty attached to a counterfactual comparison are all functionals of the same conditional second moment of the ratio-scale residual. Each level therefore accounts for that quantity exactly rather than approximately. Level 1 splits the predicted error into intrinsic and extrinsic parts by construction, and Level 2's per-input contributions are efficient with respect to the quantity being explained, whether that is the predicted error of a single run or its change between two runs, the latter being what the correction in (17) preserves when calibrated means differ. Separate off-the-shelf tools for quantification and for sensitivity analysis would not have that property. What the chain does not deliver is a single additive ladder from individual inputs up to the total, because the extrinsic term also carries the system's own irreducible noise, which no input explains; Section 4.1 is explicit about that limit. The order of the stages is also the order of the interventions available to a decision-maker: whether more computation would help, whether input quality is the binding constraint, and if so which input to attack first. Section 5.6 shows the chain changing a decision rather than only describing one.

An earlier, substantially shorter treatment of the residual-based construction appears in [17]; the present paper develops the error decomposition, hierarchical interpretation, attribution theory, validation, and counterfactual implications. The approach builds directly on the input-uncertainty literature associated with Barry L. Nelson and collaborators, which established that simulation output uncertainty contains a component due to finite replication and another due to estimated inputs, that the latter can dominate, and that input uncertainty can be apportioned among individual input models [3, 6, 4, 28, 30, 29]; later work reduced the computational burden from dedicated nested experiments toward diagnostics obtainable from a nominal simulation experiment [16]. Our intrinsic–extrinsic distinction follows that tradition, with the difference that our objective is prediction rather than estimation alone. Section 2 takes up the relationship in detail.

Contributions. The paper makes three main contributions, one per question. They are contributions to a single construction rather than three separable methods: all three operate on the same residual-based estimand, which is what allows the answers to compose instead of being reported side by side.

- **How wrong? Residual-based total prediction error.** We derive an explicit decomposition of simulation prediction error into inherent system noise, uncertainty in the simulated mean, and squared bias, while further separating bias into input-driven and fidelity-driven components. Because the true mean is unobservable, we estimate total prediction error directly from observable squared residuals and combine the resulting error estimate with a skewness-detection procedure to construct quantiles and prediction intervals (Section 3). We also argue that the ratio scale on which error is modelled is more than a transformation: it is what makes the regression feasible in a strongly seasonal, heavy-tailed system, and what makes counterfactual error prediction possible (Sections 3.2 and 5.5).
- **What kind of error? A decision-oriented uncertainty hierarchy.** After bias is suppressed by calibration, we decompose uncertainty into intrinsic Monte Carlo error and extrinsic uncertainty. The distinction is operational: intrinsic uncertainty can be reduced by additional replications, whereas extrinsic uncertainty requires improvements outside the simulation run. We then extend the hierarchy

from quantification to diagnosis by attributing the extrinsic component to individual inputs (Section 4).

- **Why? Dependence-aware attribution with empirical validation.** We attribute predicted error in the two forms a decision-maker asks for, the error of a single run decomposed across its inputs and the change in error between two runs attributed to the inputs that moved, developing an efficiency-preserving Shapley construction for the second when the two runs' calibrated means differ, and clarify the assumptions under which variance-based and mean-based attribution routes are valid. Synthetic experiments provide known attribution ground truth; production-scale backtests then test predictive calibration, the importance of extrinsic uncertainty, and counterfactual inference. In the application, Monte Carlo-only 80% intervals cover 7.7% of realized outcomes versus 77.7% for the full framework, and including extrinsic uncertainty changes the significance conclusion in 6 of 17 weeks of a counterfactual lead-time experiment (Sections 4.2.5–5.6).

Organization. Section 2 positions the framework relative to input uncertainty, simulation metamodeling, and attribution. Section 3 answers the first question, how wrong a prediction may be, by deriving the error decomposition and residual-based estimator. Section 4 answers what kind of uncertainty remains and why, through the intrinsic–extrinsic split and input attribution. Section 5 validates the components in increasing order of inferential difficulty, from settings with observed outcomes or synthetic ground truth to counterfactuals where ground truth is unavailable. Section 6 discusses the scope and limitations, and Section 7 concludes.

2 Literature Review

Uncertainty quantification in stochastic simulation has a long history, with both frequentist and Bayesian traditions [21, 24, 31]. The bulk of this literature decomposes output uncertainty into a Monte Carlo component and an input uncertainty component, and concentrates on the latter.

Early treatments of input uncertainty relied on resampling. Bootstrapping the real-world data used to fit input models gives a direct characterization of the error induced by finite samples, at the cost of nested simulation. To reduce that cost, Bayesian model averaging approaches were introduced, treating uncertainty in both the distributional family and its parameters [10, 11, 12, 13]. A parallel line replaces simulation runs with a metamodel, most often Gaussian, so that the output distribution can be characterized analytically [3, 6].

Identifying *which* input models dominate is a separate problem from quantifying the total. Early diagnostics required a sequence of additional experiments, in the worst case one per input model, each of them substantial. Ankenman and Nelson (2012) reduced this burden [4], and Song and Nelson (2016) showed that a single diagnostic experiment suffices to recover the aggregate effect of input uncertainty, the relative contribution of each input distribution, and each distribution's sensitivity to sample size [28]. Lin et al. (2015) went further, deriving both the aggregate measure and the per-input contributions from the nominal experiment an analyst would run anyway, by retaining simulation sample paths [16].

The developments above are in large part the work of Nelson and his collaborators, who established

the distinction between finite-replication uncertainty and uncertainty arising from estimated inputs and gave the first practical methods for attributing the latter to individual input models; the present paper builds on that tradition by shifting attention from uncertainty in estimating a simulation mean to uncertainty in predicting real-system outcomes, beginning with observable simulation-to-reality residuals and extending the analysis from quantification to diagnosis and counterfactual inference. The first two authors are academic descendants of Professor Nelson, and we are particularly honored to contribute this work to the special issue in his honor, and to carry that tradition into prediction and decision-making for modern, large-scale simulation systems.

Despite this depth, three obstacles block direct application at industrial scale. First, computational: methods requiring repeated or nested experiments are infeasible when a single run is expensive and the input dimension is large. Second, data quality: real-world inputs are heavy-tailed and sparse, so outlier handling becomes a first-order concern rather than a technicality. Third, and most consequential for our purposes, the literature almost uniformly assumes the simulation mean estimator is unbiased. In production systems it is not, because neither input quality nor model fidelity is adequate. Quantifying variance around a biased center does not yield a usable risk estimate, so bias calibration and uncertainty quantification have to be handled together rather than sequentially.

A further gap concerns what decision-makers actually need. Beyond a total, they need to know whether the dominant term is one they can buy down with compute or one that requires a data-quality program, and they need uncertainty estimates under counterfactual configurations that have never been observed. Neither follows from a single aggregate variance number. Shapley values from cooperative game theory provide the attribution machinery we use for the second level of our decomposition [25]; they have been applied widely in sensitivity analysis and machine learning interpretation [23, 19, 8, 29]. Our application setting throughout is retail supply chain management [2, 1].

The literature therefore provides powerful tools for measuring input uncertainty once the relevant mean and input distributions are well defined. Our setting begins one step earlier: the object available for validation is a single realized outcome, while the simulator can be biased and its inputs are estimated and dependent. The next section develops the prediction-error quantity that can be learned from those observations before asking how to decompose it.

3 Prediction Error Quantification

We use weekly inbound arrival units as a running example: the number of units received by warehouses from external suppliers in a week. The framework is metric-agnostic and the choice is expository. Suppose a simulation is executed in week t and projects inventory flows over the following N weeks; we want the prediction error of arrivals Y_{t+n} in each target week $t+n$, $n = 1, \ldots, N$.

Table 1 collects the notation used throughout the methodological development.

The center of our predictive distribution is not estimated in this paper. It is supplied by a separate calibration model, which combines the raw simulated sample mean with a time-series prediction and is materially more accurate than the raw simulated mean from an internal study. We refer to its output as the

Table 1: Summary of notation.

Symbol	Meaning
$\mathbf{F}$	True real-world input distributions
$\widehat{\mathbf{F}}$	Estimated input distributions supplied to the simulator
$\mathbf{U}_j$	Random-number stream for simulation replication j
$Y(\mathbf{F})$	Real-system outcome under the true inputs
$\mu(\mathbf{F})$	True expected system output under $\mathbf{F}$
$\widehat{\mu}(\widehat{\mathbf{F}})$	Mean produced by the practical simulator under estimated inputs
$Z_j(\widehat{\mathbf{F}})$	Output of practical simulation replication j
$\bar{Z}(\widehat{\mathbf{F}})$	Monte Carlo sample mean of the n replications
n	Number of Monte Carlo replications
σ^2	Trial-to-trial replication variance, assumed not to depend on the inputs
$\varepsilon(\mathbf{F})$	Inherent noise of the physical system about its own mean
Residuals and predicted error	
$r(\widehat{\mathbf{F}})$	Realized residual of the mean predictor in use: $Y(\mathbf{F}) - \bar{Z}(\widehat{\mathbf{F}})$ before calibration, $Y(\mathbf{F}) - \widehat{\mu}(\widehat{\mathbf{F}})$ from Section 3.2 onward
$\tilde{r}(\widehat{\mathbf{F}})$	Ratio-scale residual $r^2(\widehat{\mathbf{F}})/\widehat{\mu}^2(\widehat{\mathbf{F}})$, the regression target, Equation (9)
$\widehat{v}(\widehat{\mathbf{F}})$	Fitted ratio-scale error prediction, an estimate of $\mathbb{E}[\tilde{r}(\widehat{\mathbf{F}}) \mid \widehat{\mathbf{F}}]$
$\widehat{f}(\widehat{\mathbf{F}})$	Predicted total prediction error on the original scale, $\widehat{\mu}^2(\widehat{\mathbf{F}})\,\widehat{v}(\widehat{\mathbf{F}})$, Equation (11). A conditional *second moment*, not a variance
$\widehat{m}$	Fitted median model, used only by the skewness test of Algorithm 1
Attribution	
$\mathscr{L}$	Index set of inputs supplied to the error model
$\phi_{l,j}$	Shapley value of input l in run j, on the ratio scale
$\Phi_l,\ \Phi'_l$	Attribution of the change in predicted error to input l, before and after the efficiency rescaling of Equation (17)
Counterfactual comparison	
$Y_t^T,\ Y_t^C$	Outcome that would be realized in week t under the treatment and control configurations
Δ_t	Treatment effect $Y_t^T - Y_t^C$, the estimand of Section 5.6
$\widehat{S}_t$	Plug-in squared error of $\widehat{\Delta}_t$, Equation (23)
$\rho_{C,T}$	Assumed correlation between the two arms' prediction errors

calibrated mean. Two reasons motivate building on it rather than on the raw simulation output. An accurate center improves quantile estimates substantially whenever the underlying distribution is close to symmetric, and adopting the same center as downstream business systems avoids inconsistency between our intervals and the point forecasts those systems already publish.

3.1 Bias and Uncertainty in Stochastic Simulation

We now derive where each error source enters. Using the notation of Table 1, consider first a hypothetical simulator with perfect fidelity driven by the true input distributions $\mathbf{F}$. Its output on trial $j = 1, \ldots, n$ can be written as

$$Y_j(\mathbf{U}_j, \mathbf{F}) = \mu(\mathbf{F}) + \varepsilon_j(\mathbf{U}_j, \mathbf{F}), \tag{1}$$

where $\mathbf{U}_1, \ldots, \mathbf{U}_n$ are independent streams of uniform random numbers, $\mu(\mathbf{F}) := \mathbb{E}[Y(\mathbf{U}, \mathbf{F})]$ is the expected output under the true inputs, and the ε_j are i.i.d. and capture trial-to-trial variability. We assume

$$\mathbb{E}_{\mathbf{U}_j}[\varepsilon_j(\mathbf{U}_j, \mathbf{F})] = 0, \qquad \mathrm{Var}_{\mathbf{U}_j}[\varepsilon_j(\mathbf{U}_j, \mathbf{F})] = \sigma^2,$$

with σ^2 not depending on $\mathbf{F}$. We abbreviate $Y_j(\mathbf{U}_j, \mathbf{F})$ as $Y_j(\mathbf{F})$ and $\varepsilon_j(\mathbf{U}_j, \mathbf{F})$ as ε_j where this is unambiguous. Under perfect inputs and perfect fidelity, $Y(\mathbf{F})$ has the same distribution as a real-world observation, so we also use $Y(\mathbf{F})$ to denote the observation itself.

In practice neither $\mathbf{F}$ nor $\mu(\cdot)$ is known. Input models are estimated from data, giving $\widehat{\mathbf{F}}$, and the simulator itself is an approximation, giving $\widehat{\mu}(\cdot)$. In the supply chain setting, $\widehat{\mathbf{F}}$ includes demand forecasts and lead time predictions, while the fidelity gap between $\mu(\cdot)$ and $\widehat{\mu}(\cdot)$ originates from unmodeled workflows and the replacement of downstream fulfillment logic with a learned emulator. The simulation system is a high-fidelity, computationally expensive model that integrates several upstream planning and production systems, and it is the same system studied in [17], [22], and [18], where the emulation component is documented most fully.

We write the jth output of the practical simulator as

$$Z_j(\widehat{\mathbf{F}}) \;=\; \widehat{\mu}(\widehat{\mathbf{F}}) + \varepsilon_j, \tag{2}$$

which carries an assumption worth making explicit, since it determines the form of the variance decomposition below.

Assumption 1 (Noise invariance)**.** The replication noise of the practical simulator has the same distribution as that of the ideal simulator of (1): the ε_j are i.i.d. with $\mathbb{E}[\varepsilon_j] = 0$ and $\mathrm{Var}[\varepsilon_j] = \sigma^2$, and σ^2 does not depend on $\widehat{\mathbf{F}}$ or on the fidelity of $\widehat{\mu}(\cdot)$.

Assumption 1 says that input error and fidelity gaps shift the simulator's conditional mean without altering its trial-to-trial dispersion. It is what allows the fidelity gap to appear in the bias of (4) and nowhere in the variance of (5). A coarser simulator might plausibly exhibit a different output variance, in which case σ^2 would carry an argument and the intrinsic term of (5) would inherit a dependence on fidelity. We do not pursue that case; Section 6 returns to the modelling choice it represents.

With (2) and Assumption 1 in force, the sample mean $\bar{Z}(\widehat{\mathbf{F}}) = n^{-1}\sum_{j=1}^{n} Z_j(\widehat{\mathbf{F}})$ decomposes as

$$\bar{Z}(\widehat{\mathbf{F}}) = \widehat{\mu}(\widehat{\mathbf{F}}) + \frac{1}{n}\sum_{j=1}^{n}\varepsilon_j = \mu(\mathbf{F}) + \underbrace{\big(\widehat{\mu}(\widehat{\mathbf{F}}) - \widehat{\mu}(\mathbf{F})\big)}_{\text{input error}} + \underbrace{\big(\widehat{\mu}(\mathbf{F}) - \mu(\mathbf{F})\big)}_{\text{fidelity gap}} + \frac{1}{n}\sum_{j=1}^{n}\varepsilon_j. \tag{3}$$

The term $\widehat{\mu}(\widehat{\mathbf{F}}) = \mathbb{E}[\bar{Z}(\widehat{\mathbf{F}}) \mid \widehat{\mathbf{F}}]$ is itself a random variable, because $\widehat{\mathbf{F}}$ was estimated from one particular collection of real-world data; a different collection would yield a different value. Taking expectation with respect to $\widehat{\mathbf{F}}$ in (3) gives the bias of $\bar{Z}(\widehat{\mathbf{F}})$ relative to the true mean,

$$\mathbb{E}[\bar{Z}(\widehat{\mathbf{F}})] - \mu(\mathbf{F}) = \underbrace{\mathbb{E}_{\widehat{\mathbf{F}}}\big[\widehat{\mu}(\widehat{\mathbf{F}}) - \widehat{\mu}(\mathbf{F})\big]}_{\text{input error}} + \underbrace{\big(\widehat{\mu}(\mathbf{F}) - \mu(\mathbf{F})\big)}_{\text{fidelity gap}}, \tag{4}$$

so total bias splits cleanly into an input-driven part and a fidelity-driven part. By the law of total variance, the uncertainty of the estimator can be decomposed as follows:

$$\mathrm{Var}[\bar{Z}(\widehat{\mathbf{F}})] = \mathrm{Var}_{\widehat{\mathbf{F}}}\big[\mathbb{E}_{\mathbf{U}}[\bar{Z}(\widehat{\mathbf{F}}) \mid \widehat{\mathbf{F}}]\big] + \mathbb{E}_{\widehat{\mathbf{F}}}\big[\mathrm{Var}_{\mathbf{U}}[\bar{Z}(\widehat{\mathbf{F}}) \mid \widehat{\mathbf{F}}]\big] = \underbrace{\mathrm{Var}_{\widehat{\mathbf{F}}}[\widehat{\mu}(\widehat{\mathbf{F}})]}_{\text{extrinsic}} + \underbrace{\frac{\sigma^2}{n}}_{\text{intrinsic}}. \tag{5}$$

Equation (5) is the structural result the rest of the paper builds on. The first term, which we call *extrinsic* uncertainty, comes from the error in the estimated inputs $\widehat{\mathbf{F}}$ and is invariant to how much computation we spend. The second, *intrinsic* uncertainty, is Monte Carlo error and vanishes as $n \to \infty$. Note that fidelity gaps enter the bias in (4) but not the variance in (5). The practical asymmetry matters: intrinsic uncertainty is bought down with compute, extrinsic uncertainty only with better input modeling.

Remark 1 (Why more simulation is not enough)**.** Increasing n makes the simulator more precise *conditional on the inputs it was given*; it does not make those inputs more accurate. Thus a simulation can converge numerically while remaining uncertain as a predictor of the real system. A small Monte Carlo standard error is therefore not, by itself, evidence of a narrow prediction interval.

Remark 2 (Estimation error versus prediction error)**.** Equations (4) and (5) describe error in *estimating* the mean $\mu(\mathbf{F})$. When simulation is used to predict a future realization rather than a mean, the dispersion of $Y(\mathbf{F})$ around its own mean enters as an additional term. That term is inherent system noise: it cannot be reduced by more replications, better inputs, or higher fidelity, and any honest prediction interval must include it.

The preceding decomposition is conceptual unless its terms can be connected to observations. Because the true mean $\mu(\mathbf{F})$ is never observed, the next step replaces mean-centered quantities with a residual that is available historically.

3.2 Residual-Based Estimation of Total Error

Estimating the terms of (4) and (5) separately would be ideal and is not possible: $\mu(\mathbf{F})$ is never observed, and a realized $Y(\mathbf{F})$ is one draw rather than a mean. What is observable is the residual

$$r(\widehat{\mathbf{F}}) \;\coloneqq\; Y(\mathbf{F}) - \bar{Z}(\widehat{\mathbf{F}}),$$

and its second moment turns out to aggregate exactly the quantities we care about. Specifically, we decompose $\mathbb{E}[r^2(\widehat{\mathbf{F}})]$ as follows:

$$\begin{aligned}\mathbb{E}\big[r^2(\widehat{\mathbf{F}})\big] &= \mathbb{E}\big[(Y(\mathbf{F}) - \mu(\mathbf{F}) + \mu(\mathbf{F}) - \bar{Z}(\widehat{\mathbf{F}}))^2\big] \\ &= \underbrace{\mathrm{Var}\big(\varepsilon(\mathbf{F})\big)}_{\text{inherent noise}} + \underbrace{\mathrm{Var}\big(\bar{Z}(\widehat{\mathbf{F}})\big)}_{\text{intrinsic + extrinsic}} + \underbrace{\big(\mathbb{E}[\mu(\mathbf{F}) - \bar{Z}(\widehat{\mathbf{F}})]\big)^2}_{\text{squared bias}}. \end{aligned} \tag{6}$$

The second equality uses $\mathbb{E}[Y(\mathbf{F}) - \mu(\mathbf{F})] = 0$ together with

$$\mathrm{Cov}\big(Y(\mathbf{F}) - \mu(\mathbf{F}),\, \mu(\mathbf{F}) - \bar{Z}(\widehat{\mathbf{F}})\big) \;=\; 0, \tag{7}$$

that is, the deviation of a future realization from its own mean is uncorrelated with the simulation output used to predict it. This holds whenever the simulation is executed before the period it predicts, using only information available at that time, which is the case throughout.

Remark 3 (What is averaged, and what is conditioned on)**.** Equation (6) is an *unconditional* statement: the extrinsic term appears only because $\widehat{\mu}(\widehat{\mathbf{F}})$ varies across the input estimates $\widehat{\mathbf{F}}$ that different periods supply. Conditional on one realized $\widehat{\mathbf{F}}$ that variation is absent, and the same algebra gives

$$\mathbb{E}\big[r^2(\widehat{\mathbf{F}}) \mid \widehat{\mathbf{F}}\big] = \underbrace{\mathrm{Var}\big(\varepsilon(\mathbf{F})\big)}_{\text{inherent noise}} + \underbrace{\frac{\sigma^2}{n}}_{\text{intrinsic}} + \underbrace{\big(\mu(\mathbf{F}) - \widehat{\mu}(\widehat{\mathbf{F}})\big)^2}_{\text{squared error of the mean prediction}}, \tag{8}$$

in which input error enters as a squared error rather than as a variance. Averaging (8) over $\widehat{\mathbf{F}}$ splits that last term into $\mathrm{Var}_{\widehat{\mathbf{F}}}[\widehat{\mu}(\widehat{\mathbf{F}})]$ plus squared bias and returns (6). Both readings are used below and it matters which is which. The estimator introduced next is a *function* of $\widehat{\mathbf{F}}$, so it targets (8); because it is fitted across periods, what it learns about how error moves with the inputs is what makes the extrinsic term of (5) recoverable and what the attribution of Section 4.2 explains.

We interpret $\mathbb{E}[r^2(\widehat{\mathbf{F}})]$ as the total prediction error, containing the irreducible noise of the physical system, the estimation uncertainty of the mean prediction split as in (5), and the squared bias contributed jointly by input error and fidelity gaps. This is the quantity we estimate.

The squared-bias term in (6) is difficult to quantify because the true system mean $\mu(\mathbf{F})$ is unobserved. Moreover, when the simulator is misspecified, squared bias may dominate the variance, causing the residual second moment to mix systematic error with uncertainty and complicating variance-based attribution. We therefore apply a separate calibration model to $\bar{Z}(\widehat{\mathbf{F}})$ that combines the simulation output with a time-series

prediction. Empirical validation indicates that the calibrated predictor is approximately unbiased. Accordingly, for the remainder of the paper, we adopt the working assumption $\mathbb{E}\left[\mu(\mathbf{F}) - \bar{Z}(\widehat{\mathbf{F}})\right] = 0$. Under this assumption, the squared-bias term vanishes, so the total prediction error can be interpreted as uncertainty. This supports treating the decomposition in Section 4 as an uncertainty decomposition rather than a general error decomposition.

From this point the residual is formed against the calibrated center rather than the raw Monte Carlo mean: we write $r(\widehat{\mathbf{F}}) = Y(\mathbf{F}) - \widehat{\mu}(\widehat{\mathbf{F}})$, with $\widehat{\mu}(\widehat{\mathbf{F}})$ now denoting the calibrated predictor. The substitution leaves (6) and (8) intact, because their derivation used only $\mathbb{E}[Y(\mathbf{F}) - \mu(\mathbf{F})] = 0$ and (7), and both hold for any predictor of $\mu(\mathbf{F})$ built from information available before the period being predicted.

Remark 4 (Calibration and uncertainty play different roles)**.** Calibration is not our uncertainty estimator. Its role is to move the center of the predictive distribution close enough to the real-system mean that the second moment of the residual can be interpreted primarily as uncertainty rather than squared systematic error. The error model then learns how the remaining dispersion changes with the inputs. A strong center and a strong uncertainty model are therefore complementary.

To build the predictor $\widehat{f}(\widehat{\mathbf{F}})$ of $\mathbb{E}[r^2(\widehat{\mathbf{F}}) \mid \widehat{\mathbf{F}}]$ we regress observed squared residuals on input features using a machine learning model; the choice of learner trades predictive power against training cost and is examined in Section 5.1. Because both $Y(\mathbf{F})$ and $\widehat{\mu}(\widehat{\mathbf{F}})$ are realized at training time, we train on a scaled target rather than the raw residual,

$$\tilde{r}(\widehat{\mathbf{F}}) = r^2(\widehat{\mathbf{F}}) / \widehat{\mu}^2(\widehat{\mathbf{F}}) = \left(Y(\mathbf{F}) / \widehat{\mu}(\widehat{\mathbf{F}}) - 1\right)^2, \tag{9}$$

and recover the original scale afterwards. That the ratio is the right scale, rather than a convenient one, is argued at the end of this subsection; here we record only why the recovery is exact rather than approximate. Because $\widehat{\mu}(\widehat{\mathbf{F}})$ is a deterministic function of $\widehat{\mathbf{F}}$, it is measurable with respect to the conditioning information and factors out of the conditional expectation:

$$\mathbb{E}\left[r^2(\widehat{\mathbf{F}}) \mid \widehat{\mathbf{F}}\right] = \widehat{\mu}^2(\widehat{\mathbf{F}})\, \mathbb{E}\left[\tilde{r}(\widehat{\mathbf{F}}) \mid \widehat{\mathbf{F}}\right]. \tag{10}$$

Writing $\widehat{v}(\widehat{\mathbf{F}})$ for a regression estimate of $\mathbb{E}[\tilde{r}(\widehat{\mathbf{F}}) \mid \widehat{\mathbf{F}}]$ obtained from the scaled training targets, which we refer to below as the *ratio-scale* error prediction, we therefore set

$$\widehat{f}(\widehat{\mathbf{F}}) \;\coloneqq\; \widehat{\mu}^2(\widehat{\mathbf{F}})\, \widehat{v}(\widehat{\mathbf{F}}), \tag{11}$$

which needs only $\widehat{\mu}(\widehat{\mathbf{F}}) > 0$, satisfied for the volume metrics considered here.

The step in (10) is available only because the target is conditional on $\widehat{\mathbf{F}}$, as set out in Remark 3. The corresponding unconditional statement is false: $\widehat{\mu}(\widehat{\mathbf{F}})$ varies across periods, so $\mathbb{E}[r^2] \neq \mathbb{E}[\tilde{r}]\,\mathbb{E}[\widehat{\mu}^2]$ in general. The ratio device and the conditional formulation stand or fall together.

The ratio scale is not a variance-stabilizing convenience applied on the way to a regression; it is what makes the regression possible, and the reason also fixes the framework's scope. Each target period yields exactly one realized outcome, so the distribution of Y_{t+n} cannot be estimated from repeated observations of that period, and the obvious remedy of pooling across periods and treating them as draws from a common

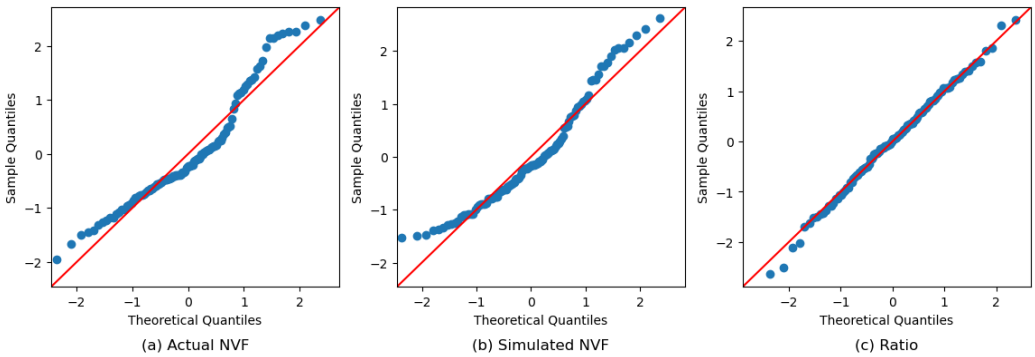


Figure 2: Normal quantile-quantile plots for a weekly aggregate series over one year of historical data. Horizontal axes are standard normal quantiles. Neither the realized outcomes nor the simulated predictions are well approximated by a single normal law across periods, and both carry a heavy right tail. The ratio of the two is much closer to the reference line

law fails badly: peak and holiday volumes differ from off-peak volumes by more than the dispersion within either regime, and the marginal distribution carries a heavy right tail, so pooling raw outcomes estimates a mixture across regimes rather than the law of any one period. Less obviously, pooling the *simulator's own output* fails for the same reason, since the simulated series inherits the seasonality and the tail of the system it models. Neither the numerator nor the denominator of (9) is, on its own, a poolable quantity.

Their ratio is. The simulator is an explicit model of the time-varying structure: it consumes the demand forecasts, lead times, and inventory positions that make one week differ from another, so dividing the realized outcome by the simulated prediction removes most of that structure and leaves a quantity approximately invariant across periods even though neither of its parts is. Two consequences carry the rest of the paper. Invariance restores a sample, because observations from different periods become draws from one distribution: the effective sample is the length of the training window rather than one, which is why Section 4.2.4 can fit an error model on roughly a hundred observations rather than on none. And the heavy right tail, being a feature of the level rather than of the relative error, does not survive the division, which licenses the parametric machinery of Section 3.3, since skewness detection against a Gaussian or skew-normal reference is reasonable on the ratio scale and would not be on the original scale.

Figure 2 shows the effect on a weekly aggregate series from this system. The three panels are normal quantile-quantile plots of, in order, the realized outcomes pooled across periods, the simulated predictions pooled the same way, and their ratio. The first two make the premise visible: both depart from normality in the same direction, with a pronounced right tail, so neither series is well described by a single law across periods. In the third the tail largely disappears, even though nothing was removed from the data except the level the simulator predicts. The panels therefore show not only that the ratio is better behaved but why. Had the heavy tail been a property of the relative error, dividing by the prediction would not have removed it, so its removal locates the tail in the level, and the level is what the simulator models. The transformation isolates the part of the discrepancy the simulator does not already explain, which is precisely the part an error model should be asked to predict.

The mechanism also marks the boundary. The argument relies on the simulator being an adequate

carrier of the time-varying structure and on the aggregate being large enough for the ratio's dispersion to be stable. Where either fails, and in practice this happens at fine granularity and for small, sparse groupings, the ratio need not be approximately invariant or approximately normal. We have observed exactly this for small, sparse aggregates, where the ratio departs from normality much as its parts do and a parametric fit degrades the mean prediction rather than improving it. In that regime the appropriate response is not to force a parametric family onto the ratio but to model its distribution nonparametrically, and the framework accommodates the substitution without any other change: the residual target, the intrinsic–extrinsic split, and the attribution machinery are all indifferent to how the ratio's distribution is represented.

3.3 Quantiles and Prediction Intervals

Predicted error and a calibrated center together determine a predictive distribution, from which we want the q-quantile $\widehat{Y}^q_{t+n}$ satisfying $\mathbb{P}(Y_{t+n} \leq \widehat{Y}^q_{t+n}) = q$ and a prediction interval $\widehat{PI}^q_{t+n}$ with $\mathbb{P}(Y_{t+n} \in \widehat{PI}^q_{t+n}) = q$. Quantiles and intervals, rather than variances, are what business stakeholders can act on: they express the range of plausible outcomes in the units of the decision.

Assuming normality is the obvious default and it fails in an instructive way. When the predicted median inherited from the calibrated mean sits off-center, coverage degrades not just at the median but across the whole quantile range, as Section 5.1 shows. We therefore test for asymmetry before committing to a distributional family. The procedure estimates the nonparametric skewness of the predictive distribution; if it departs materially from zero we fit a skew-normal, and otherwise we retain the Gaussian. Algorithm 1 summarizes the procedure. The training dataset contains K observations; for each $i = 1, \ldots, K$, x_i denotes the error-model inputs based on $\widehat{\mathbf{F}}$, and y_i denotes the corresponding realized arrivals. The predicted ratio-scale squared error v_t and median m_t for target week t are obtained from models trained on this dataset. The threshold δ governs how readily the skewed family is adopted: larger δ is more conservative, and as $\delta \to \infty$ the method reduces to the Gaussian case. We use $\delta = 0.05$, chosen empirically. Given fitted parameters, the mean and variance of the skew-normal follow from standard results [5, 14].

The test statistic $\widehat{\gamma}_t$ in Algorithm 1 is scale free, which is what makes a single threshold δ meaningful across periods of very different volume. Since $\widehat{f}_t = \widehat{\mu}_t^2 \, v_t$ by (11),

$$\frac{|m_t - \widehat{\mu}_t|}{\sqrt{\widehat{f}_t}} = \frac{|m_t/\widehat{\mu}_t - 1|}{\sqrt{v_t}},$$

so the comparison is between a relative displacement of the median and a relative dispersion, both dimensionless. Forming the ratio against $\sqrt{v_t}$ directly, without restoring the original scale first, would divide a quantity in units of the metric by a dimensionless one and would make δ volume dependent.

One caveat deserves emphasis, because it connects back to Section 3.2. Skewness detection presumes the adopted center is close to unbiased. If the calibrated mean carries appreciable bias, the procedure can mistake that bias for asymmetry and perform worse than no detection at all. The bias suppression is a precondition, not an optional refinement.

The choice of δ addresses a practical ambiguity: a small $|\widehat{\gamma}_t|$ may arise from weak asymmetry, estimation

Algorithm 1: Skewness detection for quantile and prediction interval construction

Input: Training set $\mathscr{M} = \{(x_i, y_i)\}_{i=1}^K$; calibrated means $\{\widehat{\mu}_i\}_{i=1}^K$; target week t with input x_t and calibrated mean $\widehat{\mu}_t$; skewness tolerance δ

Output: Predicted quantiles and prediction intervals for target week t

Compute scaled squared residuals $\tilde{r}_i = (y_i/\widehat{\mu}_i - 1)^2$ for $i = 1, \ldots, K$;
Form the error training set $\mathscr{R} = \{(x_i, \tilde{r}_i)\}_{i=1}^K$;
Fit the ratio-scale error model $\widehat{v}$ on $\mathscr{R}$;
Fit the ratio-scale median model $\widehat{m}$ on $\{(x_i, y_i/\widehat{\mu}_i)\}_{i=1}^K$;
Predict ratio-scale error $v_t = \widehat{v}(x_t)$ and median $m_t = \widehat{\mu}_t\,\widehat{m}(x_t)$ for target week t;
Recover the original scale, $\widehat{f}_t = \widehat{\mu}_t^2\, v_t$, as in (11);
Set $\widehat{\gamma}_t = (m_t - \widehat{\mu}_t)/\sqrt{\widehat{f}_t}$;
if $|\widehat{\gamma}_t| > \delta$ **then**
    Fit a skew-normal distribution;
    Construct quantiles and intervals from the skew-normal;
else
    Construct quantiles and intervals from the normal distribution;

noise, or both. Setting $\delta = 0$ selects a skew-normal fit for every nonzero $|\widehat{\gamma}_t|$. A positive threshold instead retains the Gaussian for small estimates, at the risk of overlooking genuine asymmetry. The analysis below quantifies how a positive threshold can shift the limiting coverage of fitted quantiles and intervals away from their nominal levels, when the fitted variance and median are consistent.

For $p \in (0,1)$, let Φ be the standard normal cumulative distribution function (CDF) and $z_p = \Phi^{-1}(p)$ its fixed p-quantile. Let $\gamma_{\max}$ be the supremum of the medians of all mean-zero, unit-variance skew-normal distributions. For any $|\eta| < \gamma_{\max}$, let Q_η be the CDF of the unique mean-zero, unit-variance skew-normal distribution with median η, and let Q_η^{-1} be its quantile function. In particular, $Q_0 = \Phi$.

For the target week t, let $\mathscr{G}_t = (x_t, \widehat{\mu}_t)$ denote the pair of error-model input and calibrated mean. The next assumption specifies the conditional skew-normal distribution of the standardized residual r_t.

Assumption 2 (Skew-normal standardized residual)**.** Define $v(x_t) := \mathbb{E}[\tilde{r}_t \mid x_t] > 0$ and $W_t := r_t/(\widehat{\mu}_t\sqrt{v(x_t)})$. Let $\gamma(x_t)$ be the median of W_t conditional on x_t. Then for $\forall w \in \mathbb{R}$,

$$\mathbb{P}(W_t \le w \mid \mathscr{G}_t) = Q_{\gamma(x_t)}(w)\ .$$

The next assumption requires the fitted error and median models to converge to their population targets as the training dataset size tends to infinity.

Assumption 3 (Consistent error and median estimators)**.** Let $\widehat{v}$ and $\widehat{m}$ in Algorithm 1 be fitted on a training dataset of size K. As $K \to \infty$, for almost every $\mathscr{G}_t$,

$$\widehat{v}(x_t) \xrightarrow{\mathbb{P}} v(x_t), \qquad \widehat{m}(x_t) \xrightarrow{\mathbb{P}} 1 + \sqrt{v(x_t)}\,\gamma(x_t)\ . \tag{12}$$

The following proposition examines how the choice of δ affects the coverage of fitted quantiles and

prediction intervals as the training dataset grows, with its proof shown in Appendix B.1.

Proposition 1 (Asymptotic coverage error from the choice of δ)**.** *Fix $q \in (0,1)$ and $\delta \geq 0$, and suppose Assumptions 2 and 3 hold. Let $\widehat{Y}_t^p$ denote the fitted p-quantile from Algorithm 1 through a training dataset with size K, and set $\widehat{PI}_t^q = [\widehat{Y}_t^{(1-q)/2}, \widehat{Y}_t^{(1+q)/2}]$. Given $\delta > 0$, assume $\mathbb{P}(|\gamma(x_t)| = \delta) = 0$. For $\gamma \in (-\gamma_{\max}, \gamma_{\max})$, define*

$$B_q(\gamma;\delta) \coloneqq \mathbf{1}\{0 < |\gamma| < \delta\}|Q_\gamma(z_q) - q|, \quad B_q^{PI}(\gamma;\delta) \coloneqq \mathbf{1}\{0 < |\gamma| < \delta\}|Q_\gamma(z_{(1+q)/2}) - Q_\gamma(-z_{(1+q)/2}) - q|.$$

where $z_p = \Phi^{-1}(p)$ is the standard-normal p-quantile and, on the Gaussian branch, the fitted distribution is normal with mean $\widehat{\mu}_t$ and variance $\widehat{f}_t = \widehat{\mu}_t^2 v_t$. Then, for almost every $\mathcal{G}_t$,

$$\lim_{K\to\infty}\left|\mathbb{P}(Y_t \leq \widehat{Y}_t^q \mid \mathcal{G}_t) - q\right| = B_q(\gamma(x_t);\delta), \quad \lim_{K\to\infty}\left|\mathbb{P}(Y_t \in \widehat{PI}_t^q \mid \mathcal{G}_t) - q\right| = B_q^{PI}(\gamma(x_t);\delta)\ .$$

Proposition 1 shows that, under the assumed skew-normal model and consistent variance and median estimates, setting $\delta = 0$ makes the conditional coverage of fitted quantiles and intervals approach their nominal levels as the dataset size $K \to \infty$. A larger δ applies the Gaussian fit to more asymmetric cases and may introduce coverage error through distributional misspecification. A positive threshold may nevertheless be useful with finite training data, because it can keep a noisy estimate of asymmetry from triggering a skew-normal fit. The choice of δ should therefore be guided by an assessment of the predictive accuracy of the fitted models.

4 Hierarchical Decomposition and Attribution

A single total-error number tells a decision-maker how much risk they carry but nothing about what to do. The two questions that follow are whether the error can be reduced at acceptable cost, and if so, which inputs to attack. Our decomposition answers them in sequence. Level 1 separates intrinsic from extrinsic uncertainty, which determines whether the answer is more compute or a data-quality program. Level 2 attributes extrinsic uncertainty across individual inputs, which determines where that program should start.

Both levels decompose the same object, the predicted error $\widehat{f}(\widehat{\mathbf{F}})$ of Section 3.2, and the sense in which each is exact differs. Level 1 is an identity: (13) defines the extrinsic term as what remains after the intrinsic part is removed, so the two sum to $\widehat{f}$ by construction. Level 2 is exact in the Shapley sense: the per-input contributions sum to whichever quantity is being explained, the predicted error of a single run or its change between two runs, which is the efficiency property that (17) restores once calibrated means differ. Neither statement says that per-input contributions sum to the extrinsic term itself, and Section 4.1 explains why that stronger statement is unavailable here.

4.1 Level 1: Intrinsic versus Extrinsic

Intrinsic uncertainty is the term σ^2/n in (5). It is directly computable from the replication sample, decays as n grows, and in a production environment can usually be reduced to negligible size for the marginal cost

of additional compute.

Everything else is then recovered as a residual,

$$\widehat{V}_{\text{ext}} \;:=\; \widehat{f}(\widehat{\mathbf{F}}) - \text{Var}_{\text{intrinsic}}, \tag{13}$$

with $\widehat{f}(\widehat{\mathbf{F}})$ the predicted total from Section 3.2. Because bias has been suppressed by calibration, this residual is uncertainty rather than a mixture of uncertainty and bias. Reducing whatever part of it is reducible at all requires improving the quality of the estimated inputs, which in an industrial setting means cross-functional forecasting initiatives on a horizon of quarters rather than a configuration change. The asymmetry in cost between the two levels is what makes the split worth reporting: it tells a stakeholder whether they are looking at a compute decision or an organizational one.

One qualification is needed, and it bears on what the split can be claimed to show. By (8), $\widehat{f}(\widehat{\mathbf{F}})$ carries inherent system noise alongside the input-driven term, and (13) removes only the intrinsic part. So $\widehat{V}_{\text{ext}}$ is the sum of the input-responsive uncertainty and the irreducible dispersion of the system itself. This is the right quantity for a prediction interval, since a realization fluctuates around its own mean whatever we do about the inputs, and Remark 2 already requires that term to be present. It is not the right quantity for sizing a data-quality program: $\widehat{V}_{\text{ext}}$ bounds from above what better inputs could remove, and the gap between that bound and the achievable reduction is precisely the inherent noise. Where the distinction matters, the two can be separated only with a direct estimate of $\text{Var}(\varepsilon(\mathbf{F}))$, which our setting does not supply. We therefore read Level 1 as separating error that additional computation can remove from error that it cannot, rather than as identifying the input-driven component on its own.

Remark 5 (Why the hierarchy is ordered)**.** Input-level attribution is useful only after the intrinsic–extrinsic split is known. If intrinsic uncertainty dominates, the immediate remedy is additional replication. If extrinsic uncertainty dominates, more computation cannot solve the problem and attribution becomes the natural next diagnostic. Level 1 therefore determines whether Level 2 is worth acting on.

4.2 Level 2: Shapley Attribution of Extrinsic Uncertainty

To apportion extrinsic uncertainty across inputs we use the Shapley value from cooperative game theory. Its logic is comparative: an input's contribution is the average change in output from adding it to a coalition of other inputs, taken over all orders in which the coalition could have formed.

Definition 1 (Shapley value)**.** For a D-player cooperative game with player set $\mathscr{D} = \{1,\dots,D\}$ and gain function $c(\cdot)$, the Shapley value of player i is

$$\phi_i = \sum_{\pi \in \Pi(\mathscr{D})} \frac{1}{D!} \big(c(P_i(\pi) \cup \{i\}) - c(P_i(\pi)) \big), \tag{14}$$

where $\Pi(\mathscr{D})$ is the set of all $D!$ orderings of $\mathscr{D}$ and $P_i(\pi)$ the players preceding i in ordering π.

Throughout the paper the players are the inputs supplied to the error model, so we identify $\mathscr{D}$ with the index set $\mathscr{L}$ of Table 1, write $D = |\mathscr{L}|$, and index inputs by $l \in \mathscr{L}$. Subsets of inputs are written $u \subseteq \mathscr{L}$.

Four properties of (14) matter here [25]. Efficiency guarantees the attributions sum to the quantity being explained, which is what makes an attribution report defensible to a stakeholder rather than merely suggestive. Symmetry gives interchangeable inputs equal credit, dummy gives irrelevant inputs none, and additivity lets attributions compose. Applied to a predictive model $p(\cdot)$, efficiency reads

$$\sum_{i \in \mathscr{D}} \phi_i = p(\mathbf{x}) - \mathbb{E}[p(\mathbf{X})]. \tag{15}$$

Attribution is asked for in two forms, and the framework supplies both. A stakeholder looking at one simulation run wants the predicted error for that run decomposed across the inputs that produced it. A stakeholder comparing two runs, or a baseline against a policy scenario, wants the *change* in predicted error attributed to the inputs that moved. Efficiency (15) specializes to each: the contributions sum to $\widehat{f}_j(\widehat{\mathbf{F}}_j)$ in the first case and to $\widehat{f}_j(\widehat{\mathbf{F}}_j) - \widehat{f}_s(\widehat{\mathbf{F}}_s)$ in the second. We develop the change form first, because it is the one in which the ratio scale creates a complication that has to be repaired, and then give the single-run form in Section 4.2.2, where no repair is needed.

Attributing change between two runs. Simulations are rerun as inputs refresh, so the same target week receives different error predictions over successive runs, and stakeholders want to know what drove the change. Let $\widehat{f}_j(\widehat{\mathbf{F}}_j)$ and $\widehat{f}_s(\widehat{\mathbf{F}}_s)$ be predicted errors from runs $j \neq s$. By (15),

$$\sum_{l \in \mathscr{L}} \phi_{l,j} = \widehat{f}_j(\widehat{\mathbf{F}}_j) - \mathbb{E}(\widehat{f}(\widehat{\mathbf{F}})), \qquad \sum_{l \in \mathscr{L}} \phi_{l,s} = \widehat{f}_s(\widehat{\mathbf{F}}_s) - \mathbb{E}(\widehat{f}(\widehat{\mathbf{F}})),$$

where the expectation is over the randomness in $\widehat{\mathbf{F}}$. Differencing,

$$\sum_{l \in \mathscr{L}} \underbrace{\left(\phi_{l,j} - \phi_{l,s}\right)}_{\Phi_l} = \widehat{f}_j(\widehat{\mathbf{F}}_j) - \widehat{f}_s(\widehat{\mathbf{F}}_s), \tag{16}$$

so Φ_l is the attribution of the change to input l.

An efficiency violation, and its correction. Equation (16) presumes a single error model $\widehat{f}$ across both runs. Two features of production practice break that presumption. Separate models are often trained per target week for business reasons, and more subtly, the model predicts error on the ratio scale of (9), so recovering error requires multiplying by $\widehat{\mu}^2(\widehat{\mathbf{F}})$ and the calibrated mean for a given target week differs across the simulation weeks being compared. Writing $\widehat{v}_j$ and $\widehat{v}_s$ for the ratio-scale error predictions of (11) in the two runs, (16) would hold as $\sum_l \phi_{l,j} - \sum_l \phi_{l,s} = \widehat{v}_j(\widehat{\mathbf{F}}_j) - \widehat{v}_s(\widehat{\mathbf{F}}_s)$ if the means agreed. When $\widehat{\mu}_j(\widehat{\mathbf{F}}_j) \neq \widehat{\mu}_s(\widehat{\mathbf{F}}_s)$,

$$\widehat{f}_j(\widehat{\mathbf{F}}_j) - \widehat{f}_s(\widehat{\mathbf{F}}_s) = \widehat{\mu}_j^2 \widehat{v}_j - \widehat{\mu}_s^2 \widehat{v}_s \;\neq\; \widehat{\mu}_j^2 \sum_{l \in \mathscr{L}} \phi_{l,j} - \widehat{\mu}_s^2 \sum_{l \in \mathscr{L}} \phi_{l,s},$$

and efficiency fails, which would make the attribution report internally inconsistent. We restore it in two steps. First we fit a single ratio-scale error model on the pooled training samples of both runs and predict the common target week with it, writing $\widehat{v}_j$ and $\widehat{v}_s$ for its two predictions. Because one model produces both,

the Shapley values $\phi_{l,j}$ and $\phi_{l,s}$ share a common reference level $\mathbb{E}(\widehat{v}(\widehat{\mathbf{F}}))$ and are efficient by construction on the ratio scale, so that

$$\sum_{l \in \mathscr{L}} \left(\phi_{l,j} - \phi_{l,s}\right) = \widehat{v}_j(\widehat{\mathbf{F}}_j) - \widehat{v}_s(\widehat{\mathbf{F}}_s).$$

Then we rescale onto the actual error change,

$$\Phi'_l = (\phi_{l,j} - \phi_{l,s}) \cdot \frac{\widehat{f}_j(\widehat{\mathbf{F}}_j) - \widehat{f}_s(\widehat{\mathbf{F}}_s)}{\widehat{v}_j(\widehat{\mathbf{F}}_j) - \widehat{v}_s(\widehat{\mathbf{F}}_s)}, \qquad \forall l \in \mathscr{L}. \tag{17}$$

Summing (17) and substituting the display above gives $\sum_{l \in \mathscr{L}} \Phi'_l = \widehat{f}_j(\widehat{\mathbf{F}}_j) - \widehat{f}_s(\widehat{\mathbf{F}}_s)$ exactly, so efficiency holds while the differing calibrated means are respected. This is what Section 1 means in claiming that the levels account for one quantity exactly: the attribution and the predicted error it explains agree by construction rather than approximately.

Remark 6 (Conditioning of the rescaling)**.** The factor in (17) divides by $\widehat{v}_j(\widehat{\mathbf{F}}_j) - \widehat{v}_s(\widehat{\mathbf{F}}_s)$, which is small whenever the two runs predict nearly the same ratio-scale error. In that regime the rescaled attributions are large in magnitude, sum to a small number, and are numerically unstable, even though efficiency still holds algebraically. The condition is diagnosable before the attribution is reported: the informative case is one in which the predicted error changed materially between runs, and when it did not, there is little change to attribute in the first place. We therefore report attributions only for comparisons in which the ratio-scale prediction moved appreciably, and we read a near-zero denominator as an indication that the question is not worth asking rather than as a result.

The single-run form is developed in Section 4.2.2; carrying the attribution from the variance scale onto the quantile scale that stakeholders actually read is developed in Section 4.2.3.

Remark 7 (Attribution is not feature importance)**.** The object being explained here is *predicted uncertainty*, or its change between runs, and in neither form is it predictive accuracy or the level of the simulated outcome. A variable can be important for predicting the mean and contribute little to uncertainty, or vice versa. Efficiency additionally requires the reported contributions to sum to the uncertainty quantity being explained; generic impurity-based importance scores do not provide that accounting identity.

4.2.1 Computing the Attribution

Equation (14) sums over $D!$ orderings, so how the values are obtained matters. Appendix A works the two-input case through in closed form, which fixes intuition for what the sum is doing before we turn to how it is computed at scale. For a black-box model they require Monte Carlo estimation, which adds sampling error on top of the error already in $\widehat{f}$. For an *additive* model class, meaning one whose members decompose as $f(\mathbf{x}) = \sum_{u \subseteq \mathscr{L}} f_u(\mathbf{x}_u)$ with f_u depending only on the coordinates in u, they are available in closed form. Functional ANOVA [15] splits the output variance as $\mathrm{Var}_{\mathbf{x}}[f(\mathbf{x})] = \sum_u \mathrm{Var}_{\mathbf{x}}[f_u(\mathbf{x}_u)] =: \sum_u R_u$, with R_u the Sobol index of the input set u, and the Shapley value of input l follows by dividing each interaction term

equally among its members, which symmetry justifies:

$$\phi_l = \sum_{u \ni l} \frac{R_u}{|u|}. \tag{18}$$

Equation (18) connects variance-based global sensitivity analysis to prediction-level attribution [23, 29]; it is what the alternative estimator of Section 4.2.5 relies on, and it is exploited with a Gaussian process emulator in [22]. Our implementation uses gradient-boosted trees, for which exact Shapley computation is likewise available [19], so no Monte Carlo estimation enters our results.

4.2.2 Attribution of a Single Run

This is the second of the two forms introduced in Section 4.2: the predicted error of a single run, decomposed across the inputs that produced it. Single-run attribution avoids the efficiency problem of the change form, since only one mean prediction is involved. Take run j. By (15), $\sum_{l \in \mathscr{L}} \phi_{l,j} = \widehat{v}_j(\widehat{\mathbf{F}}_j) - \mathbb{E}(\widehat{v}(\widehat{\mathbf{F}}))$, so the Shapley values explain the departure of predicted error from its expectation rather than the level itself. Applying the scaling factor $\alpha = \widehat{v}_j(\widehat{\mathbf{F}}_j)/(\widehat{v}_j(\widehat{\mathbf{F}}_j) - \mathbb{E}(\widehat{v}(\widehat{\mathbf{F}})))$ and returning to the original scale gives

$$\phi'_{l,j} = \alpha\, \phi_{l,j}\, \widehat{\mu}_j^2(\widehat{\mathbf{F}}_j), \qquad \forall l \in \mathscr{L},$$

which sum to the predicted error itself:

$$\sum_{l \in \mathscr{L}} \phi'_{l,j} = \alpha \Big(\widehat{v}_j(\widehat{\mathbf{F}}_j) - \mathbb{E}(\widehat{v}(\widehat{\mathbf{F}})) \Big) \widehat{\mu}_j^2(\widehat{\mathbf{F}}_j) = \widehat{v}_j(\widehat{\mathbf{F}}_j)\, \widehat{\mu}_j^2(\widehat{\mathbf{F}}_j) = \widehat{f}_j(\widehat{\mathbf{F}}_j).$$

How that total should be read deserves a word. The factor α is a normalization that moves the explained quantity from the departure of predicted error from its expectation to the predicted error itself, which is the quantity a stakeholder is shown. It does not convert the attribution into a causal statement about the whole of $\widehat{f}_j$. By (8) that total also contains the intrinsic term and the system's own inherent noise, neither of which any input explains, so the single-run contributions distribute a total that is larger than the part the inputs are responsible for. The same caution applies as in Section 4.1: the attribution ranks and compares input contributions, and it bounds rather than measures what better inputs could remove.

4.2.3 From the Variance Scale to the Quantile Scale

The construction of this subsection is derived for the change form only. It rests on factoring a difference of squares, which has no single-run counterpart, so quantile-scale attribution is available when two runs are compared and not for the level of one run. The change in predicted error is a variance-scale quantity, whereas what stakeholders read is a quantile. Closing that gap matters for the framework's stated purpose: a decomposition that explains a quantity nobody reports is of limited use. The construction below connects the two scales. Because no quantile attribution can be checked against a known value on production data, it is validated synthetically in Section 5.3.

Moving to the standard deviation scale is not a matter of taking square roots of the attributions, which

would no longer sum to anything meaningful. Instead, factor the change as a difference of squares: $\widehat{f}_j - \widehat{f}_s$ equals $(\sqrt{\widehat{f}_j} - \sqrt{\widehat{f}_s})(\sqrt{\widehat{f}_j} + \sqrt{\widehat{f}_s})$, so dividing the variance-scale attributions Φ'_l of (17) by the second factor gives

$$\varphi_l = \frac{\Phi'_l}{\sqrt{\widehat{f}_j(\widehat{\mathbf{F}}_j)} + \sqrt{\widehat{f}_s(\widehat{\mathbf{F}}_s)}}, \qquad l \in \mathscr{L}, \tag{19}$$

which sum to $\sqrt{\widehat{f}_j} - \sqrt{\widehat{f}_s}$ and so stay efficient.

The aggregate metric sums over a large number of products, so a central limit argument makes it approximately normal and a change in its τ-quantile separates into a location and a scale part,

$$v_\tau[Y_j] - v_\tau[Y_s] = \mathbb{E}[Y_j - Y_s] + z_\tau\Big(\sqrt{\widehat{f}_j(\widehat{\mathbf{F}}_j)} - \sqrt{\widehat{f}_s(\widehat{\mathbf{F}}_s)}\Big), \tag{20}$$

with z_τ the standard normal τ-quantile. Additivity of the Shapley value then allows the two parts to be attributed separately and recombined. Writing ζ_l for the attribution of the mean change, obtained by standard prediction-level attribution [19], the attribution of the τ-quantile change to input l is

$$\zeta_l + z_\tau \varphi_l, \qquad l \in \mathscr{L}, \tag{21}$$

and summing over l reproduces (20), so the quantile attribution is efficient whenever its two components are.

This closes a gap the conference version left open, where mean attribution and uncertainty attribution were separate procedures with no common account. The cost is the normal approximation, which is reasonable for aggregates over many products and would not be for a thin slice of the network.

4.2.4 Why Attribute at the Aggregate Level

Attribution in this framework operates at the aggregate reporting level: the error model of Section 3.2 is fitted on aggregate inputs against aggregate realized outcomes, and Shapley values are taken with respect to those same aggregate inputs. This matches the level at which the decisions are made. Labor and capacity commitments are set for a network group, not for an individual product, so an attribution that apportions risk across products would not be actionable even if it were more precise.

The cost is data. One aggregate observation per period means the error model trains on roughly one hundred points for a two-year window, which constrains what can be fitted and is the reason for preferring tree ensembles over higher-capacity learners in Section 5.1. It also means the residual-based construction of Section 3.2 is doing real work: with a single observation per period, sample variances are unavailable and a residual-based target is the only route to an error estimate at all.

A disaggregated alternative exists and resolves the data scarcity differently, by estimating at product level and aggregating variances upward under an independence assumption across products. That route buys many more training observations at the price of an assumption the real system violates, since shared warehouses and shared labor induce exactly the cross-product correlation that motivated event-driven simulation in Section 1. It is developed in [22] and we do not pursue it here.

4.2.5 Two Estimators and When Each Applies

There is a choice of regression response underlying everything above, and it is worth making explicit because the two options are valid under different conditions. The *variance-based* route, which is what we use, regresses the squared residual on inputs as in Section 3.2 and attributes with prediction-level Shapley values. The *mean-based* route instead fits a model of the metric itself, which the simulator already estimates without bias, and then recovers a variance attribution by decomposing the fitted function through the Sobol identity (18). The second is appealing because it avoids having to estimate an error target at all.

Stating what the mean-based route delivers requires separating a population identity, which is exact, from the estimation error incurred when a fitted model stands in for the population object. We take them in that order.

Assumption 4 (Mean-based route)**.** The pairs $(\mathbf{x}_t, y_t)$, $t = 1, \ldots, T$, are i.i.d. draws from a joint law P under which

(i) the coordinates $\{x_l\}_{l \in \mathscr{L}}$ of $\mathbf{x}$ are mutually independent, and

(ii) the conditional mean $f(\mathbf{x}) \coloneqq \mathbb{E}_P[y \mid \mathbf{x}]$ is square integrable, $\mathbb{E}_P[f^2(\mathbf{x})] < \infty$.

The following proposition establishes the exact relationship between the Sobol decomposition and Shapley attribution for the population conditional mean under Assumption 4, with its proof shown in Appendix B.2.

Proposition 2 (Sobol–Shapley identity for the population object)**.** *Under Assumption 4, f admits a functional ANOVA decomposition $f(\mathbf{x}) = \sum_{u \subseteq \mathscr{L}} f_u(\mathbf{x}_u)$ whose terms are mutually orthogonal in $L^2(P)$, so that $\mathrm{Var}_P[f(\mathbf{x})] = \sum_{u \subseteq \mathscr{L}} R_u$ with $R_u \coloneqq \mathrm{Var}_P[f_u(\mathbf{x}_u)]$. Let the cooperative game $c(u) \coloneqq \sum_{w \subseteq u} R_w$ assign to each coalition the variance it explains. Then the Shapley value* (14) *of that game satisfies*

$$\phi_l = \sum_{u \ni l} \frac{R_u}{|u|}, \qquad l \in \mathscr{L}, \tag{22}$$

which is (18)*, and $\sum_{l \in \mathscr{L}} \phi_l = \mathrm{Var}_P[f(\mathbf{x})]$.*

Two qualifications matter for how Proposition 2 is used. It concerns the population conditional mean $f = \mathbb{E}_P[y \mid \mathbf{x}]$, so in practice a fitted model stands in for f and the recovered attribution inherits whatever error that substitution carries: the identity is exact, the estimate is not. And independence enters twice, once to make the ANOVA terms orthogonal and once to make the resulting Sobol indices interpretable as separate contributions, which is why its failure is not a small perturbation but a change in what the quantities mean.

The variance-based route requires neither the independence of Assumption 4(i) nor the i.i.d. sampling that precedes it. Taking predicted error as the regression response and attributing with prediction-level Shapley values requires only the efficiency identity (15), which is an identity in conditional expectations and holds whether or not the inputs are independent. What it requires instead is an error target, which is exactly what the residual construction of Section 3.2 supplies.

Both requirements of the mean-based route fail in our setting, which is why we take the variance-based one. Independence, Assumption 4(i), fails plainly: demand, lead time, and inventory move together, so the

inputs are dependent by construction of the business rather than by accident of the data. The i.i.d. sampling premise is also strained, since aggregate observations form a weekly time series with seasonality and peak effects rather than an independent sample. Section 5.3 shows the mean-based route failing in exactly the way a violation of independence predicts, which is the empirical counterpart of this argument.

Remark 8 (Dependence is methodological, not incidental)**.** In operational systems, dependence among inputs is often created by common drivers rather than by sampling accident. Demand, lead time, inventory, and outstanding orders may all respond to seasonality, congestion, or shared planning decisions. An attribution procedure whose validity requires these features to be independently perturbable can therefore fail precisely where simulation is most useful for representing system interactions.

The methodological pieces are now in place. The numerical study validates them in a deliberate order: first predictive calibration, then the magnitude of the extrinsic component, then attribution where ground truth is known, and finally counterfactual use where direct validation is impossible.

5 Numerical Study

We evaluate the framework on a production retail supply chain simulator that models inventory flows at product granularity. The experiments form a *validation ladder*: each methodological component is first tested in the strongest setting in which ground truth or realized outcomes are available, before it is used where direct validation is weaker or impossible. Section 5.1 tests predictive calibration against realized outcomes. Section 5.2 isolates the practical consequence of omitting extrinsic uncertainty. Section 5.3 tests attribution on synthetic designs with known ground truth, after which Section 5.4 applies the same machinery to production data. Finally, Sections 5.5 and 5.6 move to counterfactual settings, where the framework must support inference without an observed treatment outcome.

Remark 9 (Scope of the study)**.** Everything reported below is an offline research study. The framework was exercised retrospectively against historical records and against constructed counterfactual scenarios; it was not deployed in, and did not control, any operating system, and no operational decision was taken on the basis of the results reported here. The counterfactual analyses of Sections 5.5 and 5.6 are semi-synthetic by construction: the control arm uses recorded historical inputs, while the treatment arm is a scenario we constructed by perturbing those inputs, so the treatment arm corresponds to no observed state of the world. Quantities are reported in relative terms throughout, as proportions or as changes expressed relative to a base; absolute volumes are withheld.

5.1 Calibration of Quantiles and Intervals

We instantiate Algorithm 1 with XGBoost [9] for the error model and quantile regression forests [20] for the median. Tree ensembles were chosen over deep models for two reasons specific to this setting: they perform better in the small-sample regime our weekly observation cadence imposes, and they expose efficient exact Shapley computation, which the attribution in Section 4.2 depends on. Our own comparisons also placed them ahead of linear regression and Gaussian process regression.

Inputs comprise starting inventory, outstanding purchase orders, key quantiles of customer demand and lead time forecasts, economic cost, and lagged ratios of realized arrivals to forecast. By *outstanding purchase orders* we mean orders already placed with manufacturers that had not been received as of the simulation start date and are expected to arrive within the horizon. These are committed inflow rather than a forecast quantity, which matters for interpreting attribution results later: an intervention on lead time can only affect orders not yet placed, so the outstanding portion is largely insulated from it. We train on two years of weekly observations and evaluate on the following year.

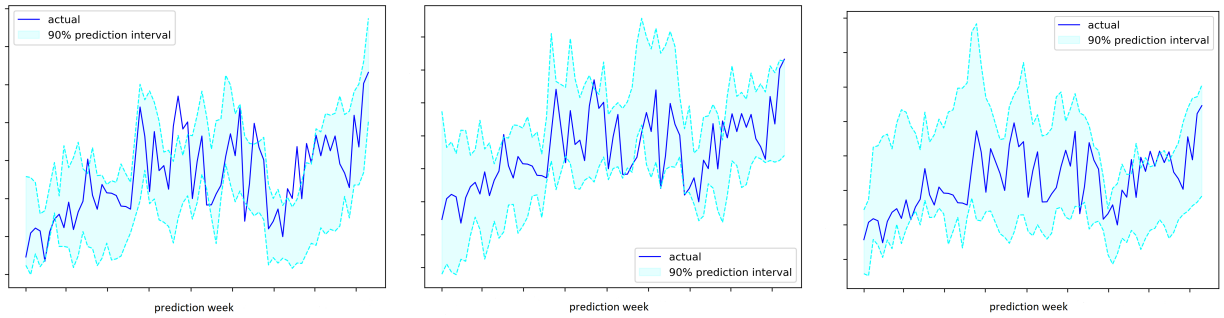
All four methods compared below operate on the ratio scale of (9), as does every result this paper reports on production data: each estimates the distribution of the ratio of realized to predicted output and recovers the original scale through (11). The synthetic designs of Section 5.3 are the exception, since their generating processes are specified directly and no simulator prediction is being divided out. They differ in two respects only, the center the distribution is placed on and how the ratio's dispersion is estimated, which is what makes the comparison informative. The ratio scale itself is not one of the variables under test; it is common to all of them, for the reasons given in Section 3.2.

Four methods are compared. MLE-RAW takes the ratio distribution to be Gaussian, estimates its parameters by maximum likelihood over the training window, and centers the resulting predictive distribution on the raw simulated mean. MLE-CALIBRATED keeps the same maximum likelihood estimate of ratio dispersion but centers on the calibrated mean. XG-CALIBRATED replaces the constant maximum likelihood dispersion with an input-dependent prediction from XGBoost, still Gaussian and still centered on the calibrated mean; this is the first method in the sequence for which the predicted error responds to the inputs at all. XG-SD-CALIBRATED adds skewness detection on top. Reading the four in order therefore isolates one change at a time: the center, then input dependence, then the distributional family. Performance is summarized by two metrics. The absolute error of empirical coverage is $|\mathscr{Q}|^{-1}\sum_{q\in\mathscr{Q}}|\widehat{q}-q|$, where $\widehat{q}$ is realized coverage at target q. The continuous ranked probability score aggregates quantile loss $\mathrm{QL}_q(\widehat{Y}^q,Y)=Y^{-1}(q(Y-\widehat{Y}^q)^+ + (1-q)(\widehat{Y}^q-Y)^+)$ as $2\sum_{q\in\mathscr{Q}}|\mathscr{D}|^{-1}\sum_{y\in\mathscr{D}}\mathrm{QL}_q(\widehat{Y}^q,y)$ over the observation set $\mathscr{D}$.

Table 2 reports six-week-ahead results. The pattern worth noting is not simply that the proposed method wins, but why the Gaussian variants lose. Under normality the calibrated mean is a poor median, which produces over-coverage concentrated at p30 and p40 and propagates outward. Skewness detection identifies exactly this condition and the resulting asymmetric fit improves both metrics, cutting coverage absolute error from 4.49% to 1.72%.

Table 2: Empirical coverage of quantile predictions for target week $t+6$. Best values in bold.

Method	p10	p20	p30	p40	p50	p60	p70	p80	p90	AE	CRPS
MLE-RAW	8.22%	27.40%	49.32%	58.90%	69.86%	78.08%	87.67%	90.41%	93.15%	12.95%	0.1152
MLE-CALIBRATED	4.11%	19.18%	36.99%	50.68%	57.53%	61.64%	65.75%	78.08%	89.04%	4.52%	0.0932
XG-CALIBRATED	13.70%	24.66%	36.99%	47.95%	57.53%	61.64%	67.12%	76.71%	91.78%	4.49%	0.0942
XG-SD-CALIBRATED	9.59%	21.92%	28.77%	39.73%	50.68%	56.16%	68.49%	76.71%	87.67%	**1.72%**	**0.0898**



(a) Target week $t+1$ (b) Target week $t+3$ (c) Target week $t+6$

Figure 3: Realized inbound arrival units against predicted means and 90% prediction intervals. Absolute values on the vertical axis are suppressed under confidentiality policy.

5.2 How Much Does Extrinsic Uncertainty Matter?

The framework's central claim is that omitting extrinsic uncertainty produces unusable risk estimates. This subsection tests that claim where it can be tested, against realized outcomes, before we rely on it in counterfactual settings where it cannot.

Setup. We construct 80% prediction intervals under two specifications. The intrinsic-only specification uses Monte Carlo sampling error alone under a normality assumption, which is what conventional practice reports. The full specification uses the proposed framework, capturing intrinsic and extrinsic components together. The evaluation covers 26 consecutive weeks on one US network group, using a 52-week rolling training window. We measure the fraction of realized observations falling inside the predicted interval.

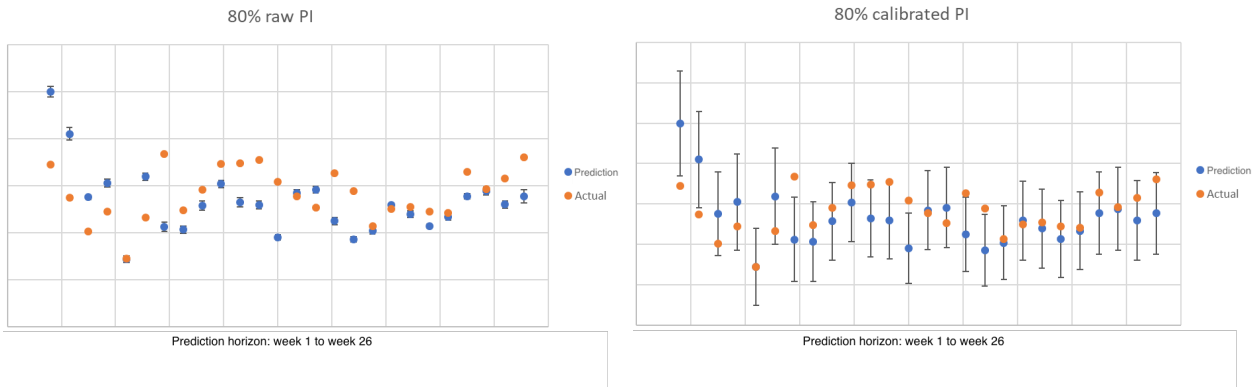
Results. Figure 4 compares the two. Intrinsic-only intervals attain 7.7% empirical coverage against an 80% target, understating risk by an order of magnitude. Adding the extrinsic component brings coverage to 77.7%, close to nominal. The gap is not a matter of degree: a method that covers 7.7% of outcomes at a nominal 80% is not a conservative risk estimate, it is a misleading one, and any decision calibrated to it inherits that error.

That extrinsic uncertainty dominates so heavily also validates the ordering of the hierarchy in Section 4. If intrinsic error were the larger term, Level 1 would be a formality and attribution would be the whole story. It is the other way round.

Having established that the extrinsic component is large enough to matter, the next question is whether its allocation across inputs can be trusted. Production data cannot reveal the correct allocation, so we move temporarily to synthetic designs where it is known by construction.

5.3 Synthetic Validation of Attribution

Attribution on production data cannot be checked, because the correct apportionment is not observable. Synthetic data removes that obstacle: if we specify the generating process, we know what the attribution



(a) Intrinsic uncertainty only

(b) Intrinsic and extrinsic uncertainty

Figure 4: Nominal 80% prediction intervals over the 26-week backtest. Empirical coverage is 7.7% under the intrinsic-only specification and 77.7% with extrinsic uncertainty included.

must be, and we can ask whether the estimator recovers it. We use two designs, chosen so that the correct answer is a set of exact zeros and any departure is unambiguous.

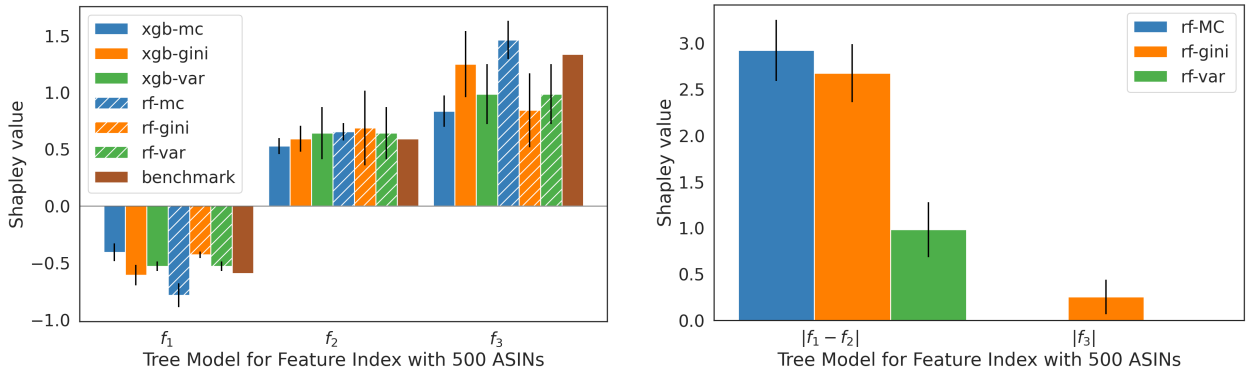
Both designs use $D = 3$ inputs, 500 observations, and 20 independent replications, with results reported as means and standard deviations across replications.

Design 1: independent inputs. Inputs are drawn independently and the response is quadratic with interaction terms, $Y = \sum_l \alpha_l X_l + \sum_{l,l'} \beta_{l,l'} X_l X_{l'} + \varepsilon$ with Gaussian noise. Input distributions differ between the two runs so that the true attribution is negative for the first input and positive for the second and third, largest for the third. Both routes of Section 4.2.5 satisfy their assumptions here, and both recover the benchmark (Figure 5(a)). This design is a control: it confirms that neither route is broken in the regime where both are licensed, so that the separation in Design 2 can be read as a consequence of the assumption failing rather than of implementation.

Design 2: dependent inputs. The response is $Y = 2X_1 + 2X_2 + \varepsilon$ with $\mathbf{x} \in \mathbb{R}^3$, so X_3 is irrelevant by construction and X_1, X_2 enter symmetrically. Inputs are drawn from a multivariate normal whose covariance makes X_1 and X_2 correlated while X_3 is independent, with observation-specific location and scale. Two quantities are then known exactly: $|\psi_1 - \psi_2| = 0$ by symmetry, and $|\psi_3| = 0$ because X_3 does not enter the response.

This design breaks the independence requirement of Assumption 4(i) and nothing else, so it isolates the cost of input dependence. Figure 5(b) reports both diagnostics. The irrelevant-input check is easy and both routes pass it, assigning approximately zero to X_3. They separate on the symmetry check: the variance-based route recovers $|\psi_1 - \psi_2| \approx 0$, while the mean-based route does not, splitting credit unequally between two inputs that are exchangeable by construction. We also include impurity-based feature importance from random forests [7] and gradient boosting [9], which are commonly pressed into service as attribution surrogates; these are less stable on both diagnostics.

The conclusion governs our production choice. Where inputs are dependent, as they are whenever



(a) Independent inputs (Design 1)

(b) Dependent inputs (Design 2)

Figure 5: Attribution against known ground truth, averaged over 20 replications with 500 observations; bars give standard deviations across replications. In (b) the true values of both reported diagnostics are zero.

demand, lead time, and inventory are driven by the same underlying conditions, the variance-based route is the one to use. The failure of the mean-based route here is not a performance gap to be tuned away, it is the independence requirement of Assumption 4(i) being false.

Design 3: recovering a quantile attribution. The two designs above validate attribution of a variance change. The construction of Section 4.2.3 makes a further claim, that a quantile change decomposes as $\zeta_l + z_\tau \varphi_l$, and a synthetic design is the natural place to test it because the generating process fixes all three targets in closed form. We reuse Design 1, whose independent inputs keep both attribution routes licensed so that any discrepancy is attributable to the recombination rather than to the estimator.

For a run with input law P, the response $Y = \sum_l \alpha_l X_l + \sum_{l,l'} \beta_{l,l'} X_l X_{l'} + \varepsilon$ has mean and variance computable from the moments of P, and under the normal reference of (20) its τ-quantile is $\mathbb{E}_P[Y] + z_\tau \sqrt{\mathrm{Var}_P[Y]}$. Changing the input law from P to P' therefore changes all three by known amounts. The true attribution of the mean change follows term by term: an additive term $\alpha_l X_l$ contributes $\alpha_l(\mathbb{E}_{P'}[X_l] - \mathbb{E}_P[X_l])$ to input l, and an interaction term $\beta_{l,l'} X_l X_{l'}$ splits between its two members exactly as in (27), each receiving its own increment weighted by the average of the other across the two laws. The true attribution of the standard deviation change follows from the same moment calculation through (19), and the true quantile attribution is then $\zeta_l + z_\tau \varphi_l$ by construction.

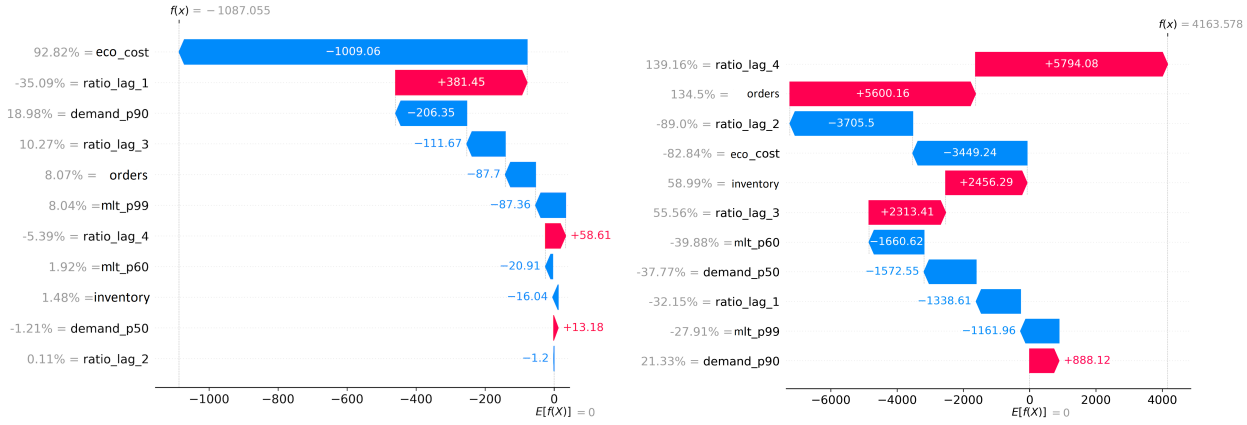
Three checks are then available against known answers: whether the estimated ζ_l recover the mean-change attribution, whether the estimated φ_l recover the standard-deviation-change attribution, and whether their combination matches the directly computed quantile attribution. The third is the one that tests (21) rather than its components, and it is the check that would be unavailable on production data, where no quantile attribution can be compared against a known value.

The synthetic study establishes which attribution route is appropriate under the dependence structure of our application. We now return to production data, where attribution cannot be checked against ground truth but can be examined for whether it recovers known operational mechanisms.

5.4 Attribution in Practice

We reuse the input set of Section 5.1 and retain XGBoost, whose exact Shapley computation makes attribution inexpensive at this scale.

Figure 6(a) explains a change in predicted error between two successive runs. Economic cost is the dominant driver of the decrease, which tracks the decline in its raw value. The lagged ratio term moves in the opposite direction: the ratio of the previous week's forecast to realized arrivals fell from 0.8 to 0.69, meaning the historical accuracy of the calibrated mean deteriorated, so the model infers a wider fidelity gap and raises predicted error. That the attribution recovers this mechanism without being told about it is the practical case for input-aware error models. Figure 6(b) shows the single-run decomposition for one target week.



(a) Change between two runs

(b) Single run

Figure 6: Uncertainty attribution by Shapley value, scaled by 10^{10}. Percentages give relative contribution.

5.5 Counterfactual Analysis: Hypothetical Economic Cost

Counterfactual analysis is where input-aware error models earn their keep, and also where evaluation is hardest, because the counterfactual outcome is by construction unobserved. We first study a setting that admits a constructed benchmark, then apply the framework where no benchmark exists at all (Section 5.6).

Simplify $Y(\mathbf{F})$ to Y and $\widehat{\mu}(\mathbf{F})$ to $\widehat{\mu}$. Let Y_u and Y_c be arrivals under unobservable and observable scenarios, with $\widehat{\mu}_u^{\text{sim}}$ and $\widehat{\mu}_c^{\text{sim}}$ the corresponding raw simulated means. The target is

$$\text{Var}(Y_u) = \text{Var}\left(\frac{Y_u}{\widehat{\mu}_u^{\text{sim}}} \cdot \widehat{\mu}_u^{\text{sim}}\right),$$

which reduces the problem to a view about the ratio $Y_u/\widehat{\mu}_u^{\text{sim}}$. Since Y_u is unobserved, an assumption is unavoidable, and we adopt the weakest one that makes evaluation possible: in each week the ratio $Y_u/\widehat{\mu}_u^{\text{sim}}$ is distributed as $Y_c/\widehat{\mu}_c^{\text{sim}}$ would be in that same week were arrivals observable. This yields a constructed benchmark

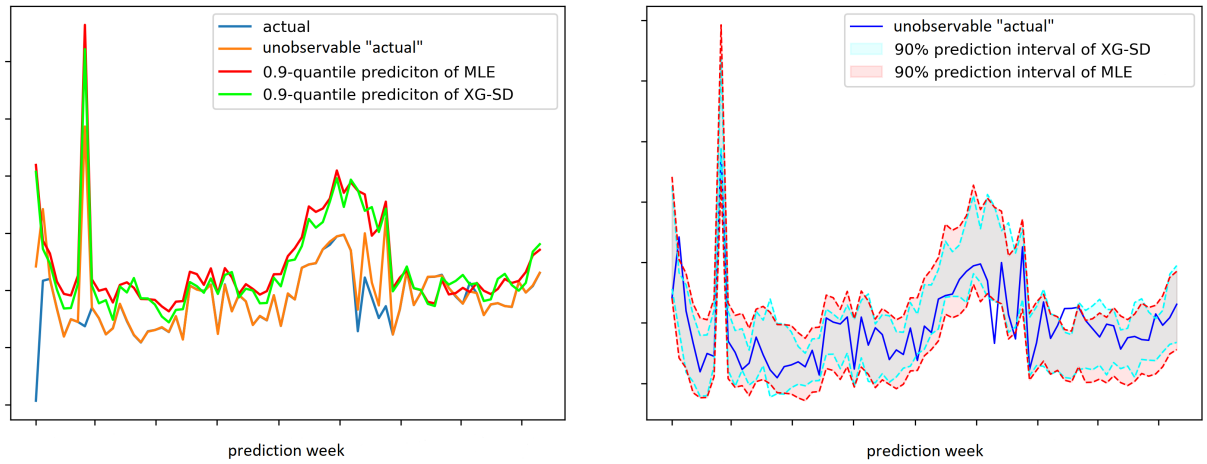
$$\widehat{Y}_u \approx \frac{Y_c}{\widehat{\mu}_c^{\text{sim}}} \cdot \widehat{\mu}_u^{\text{sim}},$$

which we treat as an evaluation device rather than as truth.

Remark 10 (Counterfactual uncertainty is an extrapolation problem)**.** A counterfactual changes not only the predicted mean but potentially the prediction error itself. Holding the historical error distribution fixed after intervening on an input therefore assumes away the mechanism the analysis is intended to measure. The input-aware error model allows uncertainty to move with the counterfactual configuration; the constructed benchmark below tests that capability where an approximate reference can be formed.

We predict the error of Y_u using the observable-period inputs with economic cost replaced by a hypothetical value, then rescale by the calibrated mean. As a benchmark we include MLE-CALIBRATED, whose ratio-scale error prediction does not respond to cost at all; the comparison isolates how much of the counterfactual capability comes from including the intervened input in the error model as opposed to the mean model.

Figure 7 and Table 3 report results with hypothetical cost set to zero against a nonzero actual. The proposed method delivers satisfactory coverage for both quantiles and intervals. MLE-CALIBRATED produces systematically larger quantiles and wider intervals, yet still responds to the cost change, because the ratio construction routes the intervention through the mean model. The practical implication is a decision rule: when the intervened input is present in the error model, use the proposed method; when it is not, the simpler MLE approach remains serviceable provided the mean model itself responds to the intervention.



(a) Quantile predictions

(b) Prediction intervals

Figure 7: Constructed unobservable arrivals against 0.9-quantile predictions (left) and 90% prediction intervals (right) for target week $t+1$.

5.6 Counterfactual Policy Experiment: Lead Time Padding Removal

Our final application is a counterfactual scenario in which no ground truth is available, which is the regime the framework is ultimately meant for. The purpose of this subsection is not to evaluate the intervention itself. It is to show that the choice of uncertainty specification determines what an analyst concludes from

Table 3: Coverage of quantiles and prediction intervals for target week $t+1$ under the unobservable scenario.

Quantile prediction	p10	p20	p30	p40	p50	p60	p70	p80	p90	AE
XG-SD-CALIBRATED	15.07%	23.29%	30.14%	45.21%	49.32%	54.79%	69.86%	82.19%	90.41%	**2.48%**
MLE-CALIBRATED	4.11%	17.81%	27.40%	38.36%	49.32%	60.27%	73.97%	83.56%	91.78%	2.51%
Prediction interval	p10	p20	p30	p40	p50	p60	p70	p80	p90	AE
XG-SD-CALIBRATED	5.48%	9.59%	27.40%	39.73%	49.32%	58.90%	65.75%	75.34%	87.67%	**3.42%**
MLE-CALIBRATED	10.96%	21.92%	38.36%	48.58%	56.16%	65.75%	76.71%	87.67%	93.15%	5.25%

the same simulation output, and therefore that the decomposition of Section 4 is decision-relevant rather than merely descriptive.

The scenario concerns padding in the manufacturer lead time (MLT) distributions consumed by the planning system. Removing such padding is expected to reduce inbound arrival units across the horizon, and the mechanism is standard inventory theory rather than a property of any particular system: lead time enters the replenishment policy through the protection interval over which demand must be covered, so shortening it lowers the target inventory position and hence the quantity ordered and eventually received [32, 26, 27]. What is not settled by theory, and what the framework is needed for, is whether the resulting change is large enough relative to its own uncertainty to support a decision. That is the question we put to it. Per Remark 9, the treatment arm is a constructed scenario, not an observed configuration.

Setup. The treatment is applied by subtracting the removed padding from each quantile of the MLT distribution. The subtracted amounts increase monotonically from the lower to the upper tail, so the intervention compresses the right tail of the lead time distribution considerably more than it shifts the center. This matters for what follows: a treatment concentrated in the tail is precisely the kind that a variance-aware analysis should treat differently from a mean shift. We predict total error for inbound arrival units under control and treatment using the demand forecast, the MLT distribution, outstanding purchase orders, and starting inventory as inputs, with two years of history for training. All inputs other than the MLT quantiles are held identical across the two arms.

Uncertainty of the treatment effect. The estimand is the difference between arms in the target metric for week t,

$$\Delta_t \;=\; Y_t^T - Y_t^C,$$

where Y_t^T and Y_t^C are the arrival units that would be realized under the treatment and control configurations. We report an interval for Δ_t itself rather than for its mean, so the relevant input is total prediction error rather than the variance of a mean estimator, and that is what $\widehat{f}$ supplies.

Let $\widehat{\mu}_t^T$ and $\widehat{\mu}_t^C$ be the calibrated centers in the two arms, with $\widehat{f}_t^T := \widehat{f}(\widehat{\mathbf{F}}^T)$ and $\widehat{f}_t^C := \widehat{f}(\widehat{\mathbf{F}}^C)$ the corresponding predicted errors, each computed as in (11). The point estimate is $\widehat{\Delta}_t = \widehat{\mu}_t^T - \widehat{\mu}_t^C$, and we combine the two error predictions by the plug-in analogue of the variance of a difference,

$$\widehat{S}_t \;:=\; \widehat{f}_t^T + \widehat{f}_t^C - 2\rho_{C,T}\sqrt{\widehat{f}_t^T\,\widehat{f}_t^C}. \tag{23}$$

We write $\widehat{S}_t$ rather than $\mathrm{Var}(\Delta_t)$ deliberately. By (8), $\widehat{f}$ is a conditional second moment that carries inherent system noise and the squared error of the mean prediction, and is therefore not the variance of the simulator's mean estimator appearing in (5). Equation (23) is an estimator we adopt, not an identity we derive.

The correlation $\rho_{C,T}$ is not identifiable from the data available. Two arguments point to a value near one. The arms share a random seed, so their Monte Carlo components move together. More importantly, only one of Y_t^T and Y_t^C can ever be realized, so the exogenous shocks that week would deliver are the same under either configuration; the inherent-noise components of the two predictions refer to one draw, not two independent ones. We therefore set $\rho_{C,T} = 1$.

That choice has a consequence worth stating, because it determines how (23) should be read. At $\rho_{C,T} = 1$ the expression factors,

$$\widehat{S}_t = \widehat{f}_t^T + \widehat{f}_t^C - 2\sqrt{\widehat{f}_t^T\,\widehat{f}_t^C} = \left(\sqrt{\widehat{f}_t^T} - \sqrt{\widehat{f}_t^C}\right)^2, \tag{24}$$

so the reported uncertainty of the treatment effect depends only on how much the intervention *changes* predicted error, not on its level. Everything common to the two arms cancels, inherent system noise included, which is the formal counterpart of the argument for $\rho_{C,T} = 1$ just given. The residual concern about inherent noise inflating the interval, raised for $\widehat{f}$ in Section 4.1, therefore does not carry over to (23): what survives the difference is the input-responsive part, which is what a treatment effect should be measured against.

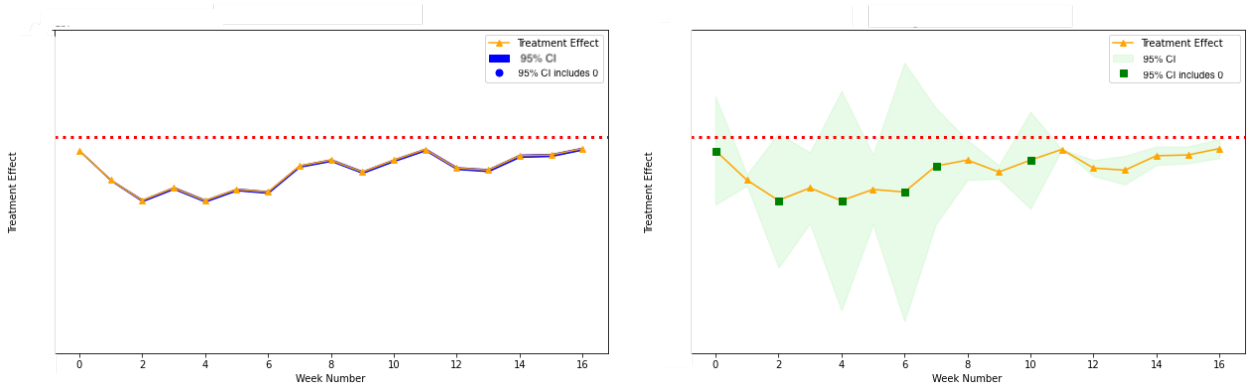
Equation (24) also links the two halves of this subsection. The width of the interval is governed by the change in predicted error between arms, and that same change is the quantity the Shapley attribution of Section 4.2 decomposes. The interval says how large the change is; the attribution says which inputs produced it. They are two readings of one number rather than two separate analyses, and we report both below.

One direction of bias remains. Setting $\rho_{C,T} = 1$ minimizes $\widehat{S}_t$ over admissible correlations and so biases the analysis *toward* declaring significance. That works against the conclusion we draw: the finding is that a substantial fraction of weeks are *not* significant once extrinsic uncertainty is included, and obtaining insignificance under an assumption that favors significance strengthens rather than weakens it. A sensitivity analysis over $\rho_{C,T} < 1$ would widen every interval and can only increase the count of insignificant weeks.

Results. Figure 8 contrasts 95% prediction intervals for Δ_t under the two specifications. With intrinsic uncertainty alone, every week in the horizon is significant: no interval contains zero. Once extrinsic uncertainty enters, 6 of the 17 simulated weeks become insignificant, and they are concentrated in the near term. In week 0 the interval is more than eight times as wide as the point estimate is large and extends well into positive territory, so neither the magnitude nor even the sign of the effect is established. Intervals exclude zero again from week 12 onward.

The point is about inference, not about the intervention. Both specifications are computed from the same simulation output and the same point estimates; they differ only in what they count as uncertainty. Yet one licenses a confident reading across the whole horizon and the other does not, which means the specification, rather than the evidence, is doing the work. An analyst who reports Monte Carlo error alone will systematically overstate what the simulation establishes, and will do so invisibly, because nothing in the output signals the omission. That is the transferable finding here, and it is a statement about how simulation

evidence should be read rather than about how any system should be configured.



(a) Intrinsic uncertainty only

(b) Intrinsic and extrinsic uncertainty

Figure 8: 95% prediction intervals for the treatment effect Δ_t by simulation week. Under the intrinsic-only specification no interval contains zero; with extrinsic uncertainty included, 6 of 17 weeks do.

Attribution. Table 4 decomposes the change in extrinsic uncertainty between arms for week 0, which fell by 24%. The p99 quantile of MLT is the largest contributor at 41%, followed by the p99 quantile of the demand forecast and the MLT median; the top three account for 75.5% of the change.

Two features of the decomposition are worth dwelling on. The first is that the demand forecast contributes appreciably even though it was held fixed across arms. This is not an artifact. Inputs act jointly through the simulation logic: shortening lead times changes the planning horizon, which changes the span of the demand forecast that the horizon consumes, and different spans carry different error. An attribution method that treated inputs as independently perturbable would miss this entirely, and a stakeholder acting on such a method would under-invest in demand forecast quality.

The second is the near-absence of outstanding purchase orders, at 1.56%, which is a useful check on the method rather than a null result. Outstanding orders are already committed inflow, placed before the simulation start; a change to the lead time distribution can only act on orders not yet placed, so this input should be close to insensitive to the intervention. That the attribution assigns it almost nothing is what one would expect if the decomposition is tracking the mechanism rather than fitting noise.

Both observations are consistency checks in this sense. Neither is a finding about the system: the couplings they recover, between lead time and the demand span a replenishment policy must cover, and between committed inflow and interventions on future orders, are properties of any lead-time-driven replenishment policy and are derived in standard treatments [32, 26]. Their value here is diagnostic. An attribution method that failed to recover relationships this well established would not be trustworthy on the relationships we cannot check independently, which is the case the method exists to handle.

Table 4: Relative contribution of inputs to the change in extrinsic uncertainty, target week 0.

Input	MLT p99	Demand p99	MLT p50	Inventory	Demand p50	Outstanding POs	MLT p10
US	40.98%	21.56%	12.96%	11.34%	11.21%	1.56%	0.38%

6 Discussion

The empirical results support a distinction that is simple mathematically but consequential operationally: precision *within* a simulator and predictive uncertainty *about* the real system are different objects. In our backtest, the intrinsic-only interval is extremely narrow relative to realized forecast error, while the full residual-based interval is close to nominal coverage. Thus additional replications can make the Monte Carlo estimate look stable without making the prediction decision-ready.

When is the extrinsic component worth estimating? Not every simulation analysis needs the full framework. If the estimand is the conditional simulation mean under a fixed set of inputs, intrinsic Monte Carlo uncertainty is the relevant quantity. Extrinsic uncertainty becomes necessary when the claim concerns a future realization, a forward-looking commitment, or a counterfactual configuration whose inputs are themselves estimated. In those settings, treating inputs as fixed answers a narrower question than the decision-maker is asking.

What does attribution add? The Level 1 decomposition identifies the *type* of uncertainty but not its source. Level 2 converts the extrinsic residual into a diagnostic: it indicates which modeled inputs account for a change in predicted uncertainty and preserves an accounting identity through Shapley efficiency. The lead-time experiment illustrates why dependence matters. An input can receive attribution even when its numerical value is held fixed across treatment and control because the intervention changes how the simulator uses that input. Attribution should therefore be interpreted through the system logic, not as independent one-variable perturbations.

Does the fidelity gap contribute variance? Our decomposition (5) places the fidelity gap in the bias term: given estimated inputs $\widehat{\mathbf{F}}$, the quantity $\widehat{\mu}(\mathbf{F}) - \mu(\mathbf{F})$ is deterministic and therefore contributes no variance. An alternative formulation writes the truth as $f^*(\mathbf{X}) + g(\mathbf{X})$, with a fidelity-gap function g evaluated at random inputs. Under that formulation $\mathrm{Var}[g(\mathbf{X})]$ is generally nonzero and fidelity contributes variance as well as bias. These are different modeling choices about where randomness resides, not contradictory empirical claims. We adopt the former throughout. Consequently, our extrinsic component should be read as the input-responsive uncertainty remaining after calibration, rather than as a separately identified fidelity variance.

Limitations. Four limitations are especially important. First, the framework suppresses bias rather than identifying it. Its interpretation therefore depends on the calibration model being close to unbiased; if calibration drifts, both the error estimate and skewness detection can deteriorate. Second, attribution is conditional

on the feature set supplied to the error model. An omitted input receives zero attribution regardless of its true influence, so the result explains the model rather than establishing a causal decomposition of the physical system. Third, the treatment-effect error measure (23) is a plug-in rather than a derived variance, and it depends on an unidentified control–treatment correlation. We set that correlation to its extreme value in the application and should report sensitivity to plausible alternatives. Fourth, the counterfactual evaluation of Section 5.5 uses a constructed benchmark; it is useful for checking whether uncertainty responds coherently to an intervention but is not observed counterfactual truth.

Generality. The framework is not tied to supply chains. It requires a stochastic simulator whose inputs are estimated, repeated historical periods for which realized outcomes eventually become observable, and a sufficiently accurate predictor of the conditional center. Patient-flow simulations, transportation and infrastructure models, service systems, and other operational digital twins can satisfy the same structure. The specific learner, input features, and distributional family can change without altering the residual-based quantification or the intrinsic–extrinsic logic.

7 Conclusion

A simulation prediction should be accompanied by answers to three questions. *How wrong might it be?* Observable residuals provide a practical target for learning total prediction error even when the true system mean is unobserved. *What kind of error is it?* Once systematic bias is suppressed by calibration, the uncertainty separates into an intrinsic Monte Carlo component and an extrinsic component that additional simulation cannot reduce. *Why does it arise?* Shapley attribution allocates that uncertainty across modeled inputs, either for a single run or as it changes between two runs, with a correction that keeps the allocation exact when calibrated means differ across the runs being compared. Because all three answers are readings of the same estimand, they compose: the split and the attribution describe the quantity the prediction interval is built from rather than a separate analysis standing beside it.

The numerical study shows why all three questions matter. In the production backtest, nominal 80% intervals based on Monte Carlo error alone cover only 7.7% of realized outcomes, whereas intervals incorporating the full predicted uncertainty cover 77.7%. Synthetic experiments show that dependence can invalidate a mean-based attribution route even when it appears reasonable under independent inputs. In the counterfactual lead-time experiment, adding extrinsic uncertainty changes the significance conclusion in 6 of 17 planning weeks. The practical message is not that every simulation requires a more complicated interval; it is that the uncertainty measure must match the claim being made about the real system.

Several extensions remain. First, a richer formulation could separate the extrinsic component into uncertainty caused by imperfect inputs and uncertainty associated with model fidelity. Doing so requires an explicit choice about whether the fidelity gap is deterministic conditional on inputs or a random function of them; the present paper adopts the former. Second, attribution is limited to features already represented in the error model, so methods that can diagnose missing drivers would connect uncertainty attribution to feature discovery. Third, aggregate estimation provides decision-level interpretability but leaves only roughly one

observation per period. Product-level estimation offers more data but requires treatment of cross-product dependence rather than assuming it away [22]. Finally, adaptive calibration and uncertainty models that track nonstationary input distributions would extend the framework to environments in which both the center and the error mechanism drift over time.

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APPENDIX

A Shapley Value Computation: A Worked Example

Because the object attributed in Section 4.2 is a *change* between two runs rather than a level, it is worth seeing the construction in a case small enough to evaluate in closed form. Take two inputs and a response $y = f(x_1, x_2)$, and attribute the move from (x_1, x_2) to (x_1', x_2'). With two players there are two orderings, and averaging the marginal contributions over them gives

$$\phi_1 = \tfrac{1}{2}\big(f(x_1', x_2) - f(x_1, x_2)\big) + \tfrac{1}{2}\big(f(x_1', x_2') - f(x_1, x_2')\big), \tag{25}$$

$$\phi_2 = \tfrac{1}{2}\big(f(x_1, x_2') - f(x_1, x_2)\big) + \tfrac{1}{2}\big(f(x_1', x_2') - f(x_1', x_2)\big). \tag{26}$$

Each input is credited with its own increment evaluated once at the other input's old value and once at its new value. Efficiency is immediate: $\phi_1 + \phi_2 = f(x_1', x_2') - f(x_1, x_2)$.

The purely interactive case is instructive. Setting $f(x_1, x_2) = x_1 x_2$ in (25)–(26) gives

$$\phi_1 = \tfrac{1}{2}(x_1' - x_1)(x_2 + x_2'), \qquad \phi_2 = \tfrac{1}{2}(x_2' - x_2)(x_1 + x_1'). \tag{27}$$

Two features of (27) carry over to the attribution reported in Section 5.6. First, an input's credit scales with its own increment but is weighted by the *average* of the other input across the two configurations, so the interaction is split symmetrically rather than assigned to whichever input moved. Second, an input whose value is genuinely unchanged receives nothing: if $x_2' = x_2$ then $\phi_2 = 0$. Nonzero attribution to an input that a policy did not act on therefore requires that the input's realized value nonetheless differed between runs, which is exactly the mechanism Section 5.6 identifies for the demand forecast, whose consumed span changes when the planning horizon changes. A closed form of this kind is available only for very small D; the general case is (14), computed exactly for the tree ensembles used here as described in Section 4.2.1.

B Proofs

B.1 Proof of Proposition 1

Proof. Assumption 2 gives the continuous conditional CDF

$$\mathbb{P}(Y_t \le y \mid \mathscr{G}_t) = Q_{\gamma(x_t)}\left(\frac{y - \widehat{\mu}_t}{\widehat{\mu}_t \sqrt{v(x_t)}}\right).$$

By (12), positivity of $\widehat{v}(x_t)$, and the continuous mapping theorem, for almost every $\mathscr{G}_t$, we have

$$v_t = \widehat{v}(x_t) \xrightarrow{\mathbb{P}} v(x_t), \qquad \widehat{\gamma}_t = \frac{\widehat{m}(x_t) - 1}{\sqrt{v_t}} \xrightarrow{\mathbb{P}} \gamma(x_t).$$

Additionally,

$$\widetilde{\gamma}_t = \widehat{\gamma}_t \mathbf{1}\{|\widehat{\gamma}_t| > \delta\} \xrightarrow{\mathbb{P}} \gamma_\delta(x_t) \coloneqq \gamma(x_t)\mathbf{1}\{|\gamma(x_t)| > \delta\}.$$

Define $\gamma_\delta(x_t) \coloneqq \gamma(x_t)\mathbf{1}\{|\gamma(x_t)| > \delta\}$. By Assumption 3 and continuity of $Q_\eta^{-1}(p)$ in η, for any fixed $p \in (0,1)$,

$$\widehat{Y}_t^p = \widehat{\mu}_t + \sqrt{\widehat{f}_t}\, Q_{\widetilde{\gamma}_t}^{-1}(p) \xrightarrow{\mathbb{P}} c_p \coloneqq \widehat{\mu}_t + \widehat{\mu}_t \sqrt{v(x_t)}\, Q_{\gamma_\delta(x_t)}^{-1}(p).$$

Slutsky's theorem gives

$$\mathbb{P}(Y_t \le \widehat{Y}_t^p \mid \mathscr{G}_t) \longrightarrow Q_{\gamma(x_t)}\Big(Q_{\gamma_\delta(x_t)}^{-1}(p)\Big). \tag{28}$$

With $p_- = (1-q)/2$ and $p_+ = (1+q)/2$, the limiting interval coverage is the difference between the limits in (28) at p_+ and p_-.

If $\gamma(x_t) = 0$ or $|\gamma(x_t)| > \delta$, then $\gamma_\delta(x_t) = \gamma(x_t)$. The limit in (28) is p, so both coverage errors are zero. If $0 < |\gamma(x_t)| < \delta$, then $\gamma_\delta(x_t) = 0$. Since $Q_0 = \Phi$ and $z_{p_-} = -z_{p_+}$, the quantile and interval coverages tend to $Q_{\gamma(x_t)}(z_q)$ and $Q_{\gamma(x_t)}(z_{p_+}) - Q_{\gamma(x_t)}(-z_{p_+})$, respectively. Their absolute differences from q are $B_q(\gamma(x_t);\delta)$ and $B_q^{PI}(\gamma(x_t);\delta)$, as claimed. □

B.2 Proof of Proposition 2

Proof. Mutual independence of the coordinates under P makes the ANOVA projections orthogonal, which gives both the decomposition of f and the additivity $\mathrm{Var}_P[f] = \sum_u R_u$; square integrability makes every term finite. Orthogonality also gives c the stated form, since a coalition u explains exactly the variance carried by the ANOVA terms it contains. For a game of that form, the marginal contribution of player l to a preceding set $P_l(\pi)$ is $\sum\{R_u : l \in u,\ u \subseteq P_l(\pi) \cup \{l\}\}$, so in (14) the term R_u is credited to l precisely when l appears last among the members of u. By symmetry among those members, that happens in a fraction $1/|u|$ of the $D!$ orderings, which gives (22). Summing over l and exchanging the order of summation, each R_u is counted $|u|$ times with weight $1/|u|$, so $\sum_{l \in \mathscr{L}} \phi_l = \sum_u R_u = \mathrm{Var}_P[f]$. The identity is due to [23]; we restate it to fix the hypotheses under which the comparison of the two routes below is exact. □